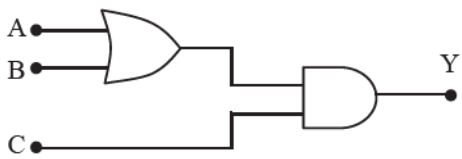
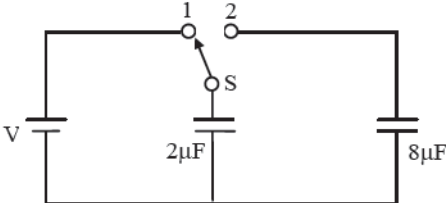


NEET (UG) - 2016 (Phase I) Question Paper

- The magnetic susceptibility is negative for
(A) Ferromagnetic material only
(B) Paramagnetic and ferromagnetic materials
(C) Diamagnetic material only
(D) Paramagnetic material only
- A uniform rope of length L and mass m_1 hangs vertically from a rigid support. A block of mass m_2 is attached to the free end of the rope. A transverse pulse of wavelength λ_1 is produced at the lower end of the rope. The wavelength of the pulse when it reaches the top of the rope is λ_2 . The ratio $\frac{\lambda_2}{\lambda_1}$ is
(A) $\sqrt{\frac{m_2}{m_1}}$ (B) $\sqrt{\frac{m_1+m_2}{m_1}}$
(C) $\sqrt{\frac{m_1}{m_2}}$ (D) $\sqrt{\frac{m_1+m_2}{m_2}}$
- The angle of incidence for a ray of light at a refracting surface of a prism is 45° . The angle of prism is 60° . If the ray suffers minimum deviation through the prism, the angle of minimum deviation and refractive index of the material of the prism respectively, are
(A) $45^\circ; \sqrt{2}$ (B) $30^\circ; \frac{1}{\sqrt{2}}$
(C) $45^\circ; \frac{1}{\sqrt{2}}$ (D) $30^\circ; \sqrt{2}$
- If the magnitude of sum of two vectors is equal to the magnitude of difference of the two vectors, the angle between these vectors is
(A) 45° (B) 180°
(C) 0° (D) 90°
- A long solenoid has 1000 turns. When a current of 4 A flows through it, the magnetic flux linked with each turn of the solenoid is 4×10^{-3} Wb. The self-inductance of the solenoid is
(A) 2 H (B) 1 H
(C) 4 H (D) 3 H
- To get an output 1 for the following circuit, the correct choice for the input is

(A) $A = 1, B = 1, C = 0$
(B) $A = 1, B = 0, C = 1$
(C) $A = 0, B = 1, C = 0$
(D) $A = 1, B = 0, C = 0$
- A uniform circular disc of radius 50 cm at rest is free to turn about an axis which is perpendicular to its plane and passes through its centre. It is subjected to a torque which produces a constant angular acceleration of 2.0 rad s^{-2} . Its net acceleration in ms^{-2} at the end of 2.0 s is approximately
(A) 6.0 (B) 3.0
(C) 8.0 (D) 7.0
- A particle of mass 10 g moves along a circle of radius 6.4 cm with a constant tangential acceleration. What is the magnitude of this acceleration, if the kinetic energy of the particle becomes equal to 8×10^{-4} J by the end of the second revolution after the beginning of the motion?
(A) 0.18 m/s^2 (B) 0.2 m/s^2
(C) 0.1 m/s^2 (D) 0.15 m/s^2
- An inductor 20 mH, a capacitor 50 μF and a resistor 40 Ω are connected in series across a source of emf $V = 10 \sin 340t$. The power loss in A.C. circuit is
(A) 0.76 W (B) 0.89 W
(C) 0.51 W (D) 0.67 W
- Two non-mixing liquids of densities ρ and $n\rho$ ($n > 1$) are put in a container. The height of each liquid is h . A solid cylinder of length L and density d is put in this container. The cylinder floats with its axis vertical and length pL ($p < 1$) in the denser liquid. The density d is equal to
(A) $[2 + (n-1)p]\rho$
(B) $[1 + (n-1)p]\rho$
(C) $[1 + (n+1)p]\rho$
(D) $[2 + (n+1)p]\rho$
- A refrigerator works between 4°C and 30°C . It is required to remove 600 calories of heat every second in order to keep the temperature of the refrigerated space constant. The power required is [Take 1 cal = 4.2 joules]
(A) 236.5 W (B) 2365 W
(C) 2.365 W (D) 23.65 W
- A body of mass 1 kg begins to move under the action of a time dependent force $F = (2t \hat{i} + 3t^2 \hat{j})\text{N}$, where $\hat{i}$ and $\hat{j}$ are unit vectors along x and y axis. What power will be developed by the force at the time t ?
(A) $(2t^3 + 3t^4)\text{W}$ (B) $(2t^3 + 3t^5)\text{W}$
(C) $(2t^2 + 3t^2)\text{W}$ (D) $(2t^2 + 4t^4)\text{W}$



13. From a disc of radius R and mass M , a circular hole of diameter R , whose rim passes through the centre is cut. What is the moment of inertia of the remaining part of the disc about a perpendicular axis, passing through the centre?
- (A) $11 MR^2/32$ (B) $9 MR^2/32$
(C) $15 MR^2/32$ (D) $13 MR^2/32$
14. A particle moves so that its position vector is given by $\vec{r} = \cos\omega t \hat{x} + \sin\omega t \hat{y}$, where ω is a constant. Which of the following is true?
- (A) Velocity is perpendicular to $\vec{r}$ and acceleration is directed towards the origin.
(B) Velocity is perpendicular to $\vec{r}$ and acceleration is directed away from the origin.
(C) Velocity and acceleration both are perpendicular to $\vec{r}$.
(D) Velocity and acceleration both are parallel to $\vec{r}$.
15. The ratio of escape velocity at earth (v_e) to the escape velocity at a planet (v_p) whose radius and mean density are twice as that of earth is
- (A) $1 : 4$ (B) $1 : \sqrt{2}$
(C) $1 : 2$ (D) $1 : 2\sqrt{2}$
16. Coefficient of linear expansion of brass and steel rods are α_1 and α_2 . Lengths of brass and steel rods are l_1 and l_2 respectively. If $(l_2 - l_1)$ is maintained same at all temperatures, which one of the following relations holds good?
- (A) $\alpha_1^2 l_2 = \alpha_2^2 l_1$ (B) $\alpha_1 l_1 = \alpha_2 l_2$
(C) $\alpha_1 l_2 = \alpha_2 l_1$ (D) $\alpha_1 l_2^2 = \alpha_2 l_1^2$
17. A piece of ice falls from a height h so that it melts completely. Only one-quarter of the heat produced is absorbed by the ice and all energy of ice gets converted into heat during its fall. The value of h is
(Latent heat of ice is 3.4×10^5 J/kg and $g = 10$ N/kg)
- (A) 136 km (B) 68 km
(C) 34 km (D) 544 km
18. A disc and a sphere of same radius but different masses roll off on two inclined planes of the same altitude and length. Which one of the two objects gets to the bottom of the plane first?
- (A) Both reach at the same time
(B) Depends on their masses
(C) Disc
(D) Sphere
19. A gas is compressed isothermally to half its initial volume. The same gas is compressed separately through an adiabatic process until its volume is again reduced to half. Then
- (A) Compressing the gas isothermally or adiabatically will require the same amount of work.
(B) Which of the case (whether compression through isothermal or through adiabatic process) requires more work will depend upon the atomicity of the gas.
(C) Compressing the gas isothermally will require more work to be done.
(D) Compressing the gas through adiabatic process will require more work to be done.
20. An air column, closed at one end and open at the other, resonates with a tuning fork when the smallest length of the column is 50 cm. The next larger length of the column resonating with the same tuning fork is
- (A) 150 cm (B) 200 cm
(C) 66.7 cm (D) 100 cm
21. Two identical charged spheres suspended from a common point by two massless strings of lengths l , are initially at a distance d ($d \ll l$) apart because of their mutual repulsion. The charges begin to leak from both the spheres at a constant rate. As a result, the spheres approach each other with a velocity v . Then v varies as a function of the distance x between the spheres, as
- (A) $v \propto x^{-\frac{1}{2}}$ (B) $v \propto x^{-1}$
(C) $v \propto x^{\frac{1}{2}}$ (D) $v \propto x$
22. A capacitor of $2 \mu\text{F}$ is charged as shown in the diagram. When the switch S is turned to position 2, the percentage of its storage energy dissipated is
- 
- (A) 75% (B) 80%
(C) 0% (D) 20%
23. The charge flowing through a resistance R varies with time t as $Q = at - bt^2$, where a and b are positive constants. The total heat produced in R is
- (A) $\frac{a^3 R}{2b}$ (B) $\frac{a^3 R}{b}$
(C) $\frac{a^3 R}{6b}$ (D) $\frac{a^3 R}{3b}$



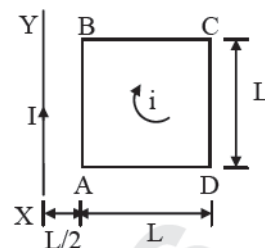
24. Match the corresponding entries of column-I with column-II (Where m is the magnification produced by the mirror)

	Column - I		Column - II
i.	$m = -2$	a.	Convex mirror
ii.	$m = -\frac{1}{2}$	b.	Concave mirror
iii.	$m = +2$	c.	Real image
iv.	$m = +\frac{1}{2}$	d.	Virtual image

- (A) $i \rightarrow a$ and d ; $ii \rightarrow b$ and c ; $iii \rightarrow b$ and d ; $iv \rightarrow b$ and c
 (B) $i \rightarrow c$ and d ; $ii \rightarrow b$ and d ; $iii \rightarrow b$ and c ; $iv \rightarrow a$ and d
 (C) $i \rightarrow b$ and c ; $ii \rightarrow b$ and c ; $iii \rightarrow b$ and d ; $iv \rightarrow a$ and d
 (D) $i \rightarrow a$ and c ; $ii \rightarrow a$ and d ; $iii \rightarrow a$ and b ; $iv \rightarrow c$ and d
25. A potentiometer wire is 100 cm long and a constant potential difference is maintained across it. Two cells are connected in series' first to support one another and then in opposite direction. The balance points are obtained at 50 cm and 10 cm from the positive end of the wire in the two cases. The ratio of emf's is
 (A) 3 : 4 (B) 3 : 2
 (C) 5 : 1 (D) 5 : 4
26. The molecules of a given mass of a gas have r.m.s velocity of 200 ms^{-1} at 27°C and $1.0 \times 10^5 \text{ Nm}^{-2}$ pressure. When the temperature and pressure of the gas are respectively, 127°C and $0.05 \times 10^5 \text{ Nm}^{-2}$, the r.m.s. velocity of its molecules in ms^{-1} is
 (A) $\frac{100\sqrt{2}}{3}$ (B) $\frac{100}{3}$
 (C) $100\sqrt{2}$ (D) $\frac{400}{\sqrt{3}}$
27. A long straight wire of radius a carries a steady current I . The current is uniformly distributed over its cross-section. The ratio of the magnetic fields B and B' at radial distances $\frac{a}{2}$ and $2a$ respectively, from the axis of the wire is
 (A) 1 (B) 4 (C) $\frac{1}{4}$ (D) $\frac{1}{2}$
28. A small signal voltage $V(t) = V_0 \sin \omega t$ is applied across an ideal capacitor C
 (A) Current $I(t)$ is in phase with voltage $V(t)$.
 (B) Current $I(t)$ leads voltage $V(t)$ by 180° .
 (C) Current $I(t)$ lags voltage $V(t)$ by 90° .
 (D) Over a full cycle the capacitor C does not consume any energy from the voltage source.

29. A square loop ABCD carrying a current i , is placed near and coplanar with a long straight conductor XY carrying a current I , the net force on the loop will be

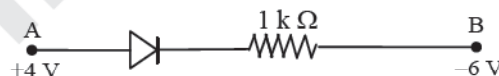
(A) $\frac{2\mu_0 i I L}{3\pi}$
 (B) $\frac{\mu_0 i I L}{2\pi}$
 (C) $\frac{2\mu_0 i I}{3\pi}$
 (D) $\frac{\mu_0 i I}{2\pi}$



30. An electron of mass m and a photon have same energy E . The ratio of de-Broglie wavelengths associated with them is (c being velocity of light)

(A) $c(2mE)^{\frac{1}{2}}$ (B) $\frac{1}{c} \left(\frac{2m}{E} \right)^{\frac{1}{2}}$
 (C) $\frac{1}{c} \left(\frac{E}{2m} \right)^{\frac{1}{2}}$ (D) $\left(\frac{E}{2m} \right)^{\frac{1}{2}}$

31. Consider the junction diode as ideal. The value of current flowing through AB is



- (A) 10^{-1} A (B) 10^{-3} A
 (C) 0 A (D) 10^{-2} A
32. Out of the following options which one can be used to produce a propagating electromagnetic wave
 (A) A chargeless particle
 (B) An accelerating charge
 (C) A charge moving at constant velocity
 (D) A stationary charge
33. A siren emitting a sound of frequency 800 Hz moves away from an observer towards a cliff at a speed of 15 ms^{-1} . Then, the frequency of sound that the observer hears in the echo reflected from the cliff is (Take velocity of sound in air = 330 ms^{-1})
 (A) 838 Hz (B) 885 Hz
 (C) 765 Hz (D) 800 Hz
34. The intensity at the maximum in a Young's double slit experiment is I_0 . Distance between two slits is $d = 5\lambda$, where λ is the wavelength of light used in the experiment. What will be the intensity in front of one of the slits on the screen placed at a distance, $D = 10 d$?
 (A) $\frac{3}{4} I_0$ (B) $\frac{I_0}{2}$
 (C) I_0 (D) $\frac{I_0}{4}$



35. When a metallic surface is illuminated with radiation of wavelength λ , the stopping potential is V . If the same surface is illuminated with radiation of wavelength 2λ , the stopping potential is $\frac{V}{4}$. The threshold wavelength for metallic surface is
(A) $\frac{5}{2}\lambda$ (B) 3λ (C) 4λ (D) 5λ
36. A black body is at a temperature of 5760 K. The energy of radiation emitted by the body at wavelength 250 nm is U_1 , at wavelength 500 nm is U_2 and that at 1000 nm is U_3 . Wien's constant, $b = 2.88 \times 10^6$ nmK. Which of the following is correct?
(A) $U_1 > U_2$ (B) $U_2 > U_1$
(C) $U_1 = 0$ (D) $U_3 = 0$
37. When an α -particle of mass m moving with velocity v bombards on heavy nucleus of charge ' Ze ', its distance of closest approach from the nucleus depends on m as
(A) $\frac{1}{m^2}$ (B) m
(C) $\frac{1}{m}$ (D) $\frac{1}{\sqrt{m}}$
38. A car is negotiating a curved road of radius R . The road is banked at an angle θ . The coefficient of friction between the tyres of the car and the road is μ_s . The maximum safe velocity on this road is
(A) $\sqrt{\frac{g}{R} \frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}}$ (B) $\sqrt{\frac{g}{R^2} \frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}}$
(C) $\sqrt{gR^2 \frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}}$ (D) $\sqrt{gR \frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}}$
39. Given the value of Rydberg constant is 10^7 m^{-1} , the wave number of the last line of the Balmer series in hydrogen spectrum will be
(A) $0.25 \times 10^7 \text{ m}^{-1}$ (B) $2.5 \times 10^7 \text{ m}^{-1}$
(C) $0.025 \times 10^4 \text{ m}^{-1}$ (D) $0.5 \times 10^7 \text{ m}^{-1}$
40. What is the minimum velocity with which a body of mass m must enter a vertical loop of radius R so that it can complete the loop?
(A) $\sqrt{3gR}$ (B) $\sqrt{5gR}$
(C) $\sqrt{gR}$ (D) $\sqrt{2gR}$
41. A npn transistor is connected in common emitter configuration in a given amplifier. A load resistance of 800Ω is connected in the collector circuit and the voltage drop across it is 0.8 V . If the current amplification factor is 0.96 and the input resistance of the circuit is 192Ω , the voltage gain and the power gain of the amplifier will respectively be
(A) 4, 4 (B) 4, 3.69
(C) 4, 3.84 (D) 3.69, 3.84
42. If the velocity of a particle is $v = At + Bt^2$, where A and B are constants, then the distance travelled by it between 1 s and 2 s is
(A) $\frac{3}{2}A + \frac{7}{3}B$ (B) $\frac{A}{2} + \frac{B}{3}$
(C) $\frac{3}{2}A + 4B$ (D) $3A + 7B$
43. In a diffraction pattern due to a single slit of width a , the first minimum is observed at an angle 30° when light of wavelength $5000 \text{ \AA}$ is incident on the slit. The first secondary maximum is observed at an angle of
(A) $\sin^{-1}\left(\frac{1}{2}\right)$ (B) $\sin^{-1}\left(\frac{3}{4}\right)$
(C) $\sin^{-1}\left(\frac{1}{4}\right)$ (D) $\sin^{-1}\left(\frac{2}{3}\right)$
44. At what height from the surface of earth the gravitation potential and the value of g are $-5.4 \times 10^7 \text{ J kg}^{-2}$ and 6.0 ms^{-2} respectively? Take the radius of earth as 6400 km
(A) 1400 km (B) 2000 km
(C) 2600 km (D) 1600 km
45. A astronomical telescope has objective and eyepiece of focal length 40 cm and 4 cm respectively. To view an object 200 cm away from the objective, the lenses must be separated by a distance
(A) 50.0 cm (B) 54.0 cm
(C) 37.3 cm (D) 46.0 cm
46. Two electrons occupying the same orbital are distinguished by _____.
(A) azimuthal quantum number
(B) spin quantum number
(C) principal quantum number
(D) magnetic quantum number
47. The CORRECT statement regarding the comparison of staggered and eclipsed conformations of ethane is _____.
(A) The eclipsed conformation of ethane is more stable than staggered conformation even though the eclipsed conformation has torsional strain.
(B) The staggered conformation of ethane is more stable than eclipsed conformation, because staggered conformation has no torsional strain.
(C) The staggered conformation of ethane is less stable than eclipsed conformation, because staggered conformation has torsional strain.
(D) The eclipsed conformation of ethane is more stable than staggered conformation, because eclipsed conformation has no torsional strain.



48. When copper is heated with conc. HNO_3 , it produces _____.
(A) $\text{Cu}(\text{NO}_3)_2$, NO and NO_2
(B) $\text{Cu}(\text{NO}_3)_2$ and N_2O
(C) $\text{Cu}(\text{NO}_3)_2$ and NO_2
(D) $\text{Cu}(\text{NO}_3)_2$ and NO
49. Which of the following statements about the composition of the vapour over an ideal 1 : 1 molar mixture of benzene and toluene is CORRECT? Assume that the temperature is constant at 25°C .
(Given: Vapour pressure data at 25°C , benzene = 12.8 kPa, toluene = 3.85 kPa)
(A) The vapour will contain equal amounts of benzene and toluene.
(B) Not enough information is given to make a prediction.
(C) The vapour will contain a higher percentage of benzene.
(D) The vapour will contain a higher percentage of toluene.
50. In a protein molecule, various amino acids are linked together by _____.
(A) peptide bond (B) dative bond
(C) α -glycosidic bond (D) β -glycosidic bond
51. MY and NY_3 , two nearly insoluble salts, have the same K_{sp} values of 6.2×10^{-13} at room temperature. Which statement would be TRUE in regard to MY and NY_3 ?
(A) The salts MY and NY_3 are more soluble in 0.5 M KY than in pure water.
(B) The addition of the salt of KY to solution of MY and NY_3 will have no effect on their solubilities.
(C) The molar solubilities of MY and NY_3 in water are identical.
(D) The molar solubility of MY in water is less than that of NY_3 .
52. The CORRECT statement regarding the basicity of arylamines is _____.
(A) arylamines are generally more basic than alkylamines because of aryl group
(B) arylamines are generally more basic than alkylamines because the nitrogen atom in arylamines is sp -hybridized
(C) arylamines are generally less basic than alkylamines because the nitrogen lone-pair electrons are delocalised by interaction with the aromatic ring π electron system
(D) arylamines are generally more basic than alkylamines because the nitrogen lone-pair electrons are not delocalised by interaction with the aromatic ring π electron system
53. The electronic configurations of Eu (atomic no. 63), Gd (atomic no. 64) and Tb (atomic no. 65) are _____, respectively.
(A) $[\text{Xe}] 4f^6 5d^1 6s^2$, $[\text{Xe}] 4f^7 5d^1 6s^2$ and $[\text{Xe}] 4f^8 5d^1 6s^2$
(B) $[\text{Xe}] 4f^7 6s^2$, $[\text{Xe}] 4f^7 5d^1 6s^2$ and $[\text{Xe}] 4f^8 6s^2$
(C) $[\text{Xe}] 4f^7 6s^2$, $[\text{Xe}] 4f^8 6s^2$ and $[\text{Xe}] 4f^8 5d^1 6s^2$
(D) $[\text{Xe}] 4f^6 5d^1 6s^2$, $[\text{Xe}] 4f^7 5d^1 6s^2$ and $[\text{Xe}] 4f^9 6s^2$
54. The product obtained as a result of a reaction of nitrogen with CaC_2 is _____.
(A) CaCN_3 (B) Ca_2CN
(C) CaCN_2 (D) CaCN
55. The addition of a catalyst during a chemical reaction alters which of the following quantities?
(A) Entropy
(B) Internal energy
(C) Enthalpy
(D) Activation energy
56. Predict the CORRECT order among the following.
(A) bond pair – bond pair > lone pair – bond pair > lone pair – lone pair
(B) lone pair – bond pair > bond pair – bond pair > lone pair – lone pair
(C) lone pair – lone pair > lone pair – bond pair > bond pair – bond pair
(D) lone pair – lone pair > bond pair – bond pair > lone pair – bond pair
57. The pair of electrons in the given carbanion, $\text{CH}_3\text{C} \equiv \text{C}^-$ is present in which of the following orbitals?
(A) sp^2 (B) sp
(C) $2p$ (D) sp^3
58. The product formed by the reaction of an aldehyde with a primary amine is _____.
(A) carboxylic acid (B) aromatic acid
(C) Schiff's base (D) ketone
59. Which one given below is a non-reducing sugar?
(A) Glucose (B) Sucrose
(C) Maltose (D) Lactose
60. The pressure of H_2 required to make the potential of H_2 electrode zero in pure water at 298 K is _____.
(A) 10^{-10} atm (B) 10^{-4} atm
(C) 10^{-14} atm (D) 10^{-12} atm
61. Which of the following has the longest C – O bond length? (Free C – O bond length in CO is 1.128 Å)
(A) $[\text{Fe}(\text{CO})_4]^{2-}$ (B) $[\text{Mn}(\text{CO})_6]^+$
(C) $\text{Ni}(\text{CO})_4$ (D) $[\text{Co}(\text{CO})_4]^-$



62. Consider the molecules CH_4 , NH_3 and H_2O . Which of the given statements is FALSE?

- (A) The $\text{H}-\text{O}-\text{H}$ bond angle in H_2O is smaller than the $\text{H}-\text{N}-\text{H}$ bond angle in NH_3 .
 (B) The $\text{H}-\text{C}-\text{H}$ bond angle in CH_4 is larger than the $\text{H}-\text{N}-\text{H}$ bond angle in NH_3 .
 (C) The $\text{H}-\text{C}-\text{H}$ bond angle in CH_4 , the $\text{H}-\text{N}-\text{H}$ bond angle in NH_3 , and the $\text{H}-\text{O}-\text{H}$ bond angle in H_2O are all greater than 90° .
 (D) The $\text{H}-\text{O}-\text{H}$ bond angle in H_2O is larger than the $\text{H}-\text{C}-\text{H}$ bond angle in CH_4 .

63. The rate of a first-order reaction is $0.04 \text{ mol L}^{-1} \text{ s}^{-1}$ at 10 seconds and $0.03 \text{ mol L}^{-1} \text{ s}^{-1}$ at 20 seconds after initiation of the reaction. The half-life period of the reaction is _____.

- (A) 44.1 s (B) 54.1 s
 (C) 24.1 s (D) 34.1 s

64. Fog is a colloidal solution of _____.

- (A) solid in gas (B) gas in gas
 (C) liquid in gas (D) gas in liquid

65. In which of the following options the order of arrangement does not agree with the variation of property indicate against it ?

- (A) $\text{B} < \text{C} < \text{N} < \text{O}$ (increasing first ionisation enthalpy)
 (B) $\text{I} < \text{Br} < \text{Cl} < \text{F}$ (increasing electron gain enthalpy)
 (C) $\text{Li} < \text{Na} < \text{K} < \text{Rb}$ (increasing metallic radius)
 (D) $\text{Al}^{3+} > \text{Mg}^{2+} < \text{Na}^+ < \text{F}$ (increasing ionic size)

66. Which of the following statements is INCORRECT?

- (A) Ca^{2+} ions are not important in maintaining the regular beating of the heart.
 (B) Mg^{2+} ions are important in the green parts of the plants.
 (C) Mg^{2+} ions form a complex with ATP.
 (D) Ca^{2+} ions are important in blood clotting.

67. The CORRECT thermodynamic conditions for the spontaneous reaction at all temperatures is _____.

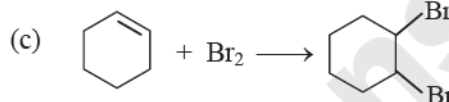
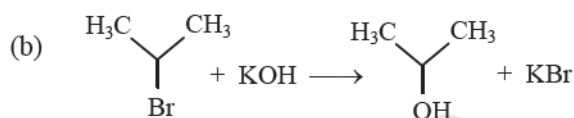
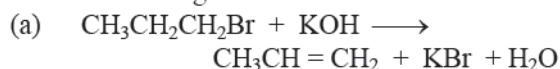
- (A) $\Delta H < 0$ and $\Delta S > 0$
 (B) $\Delta H < 0$ and $\Delta S < 0$
 (C) $\Delta H < 0$ and $\Delta S = 0$
 (D) $\Delta H > 0$ and $\Delta S < 0$

68. Match the compounds given in column-I with the hybridisation and shapes given in column-II and mark the CORRECT option.

	Column I		Column II
i.	XeF_6	a.	Distorted octahedral
ii.	XeO_3	b.	Square planar
iii.	XeOF_4	c.	Pyramidal
iv.	XeF_4	d.	Square pyramidal

- (A) i - d, ii - c, iii - a, iv - b
 (B) i - d, ii - a, iii - b, iv - c
 (C) i - a, ii - c, iii - d, iv - b
 (D) i - a, ii - b, iii - d, iv - c

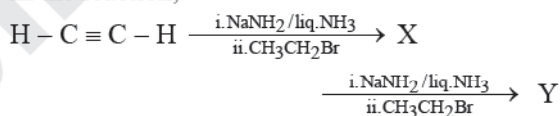
69. For the following reactions:



which of the following statements is CORRECT?

- (A) (a) is elimination, (b) and (c) are substitution reactions.
 (B) (a) is substitution, (b) and (c) are addition reactions.
 (C) (a) and (b) are elimination reactions and (c) is addition reaction.
 (D) (a) is elimination, (b) is substitution and (c) is addition reaction.

70. In the reaction,



X and Y are _____.

- (A) X = But-2-yne ; Y = Hex-2-yne
 (B) X = But-1-yne ; Y = Hex-2-yne
 (C) X = But-1-yne ; Y = Hex-3-yne
 (D) X = But-2-yne ; Y = Hex-3-yne

71. Equal moles of hydrogen and oxygen gases are placed in a container with a pin-hole through which both can escape. What fraction of the oxygen escapes in the time required for one-half of the hydrogen to escape?

- (A) $\frac{3}{8}$ (B) $\frac{1}{2}$ (C) $\frac{1}{8}$ (D) $\frac{1}{4}$

72. Which is the CORRECT statement for the given acids?

- (A) Phosphinic acid is a monoprotic acid while phosphonic acid is a diprotic acid.
 (B) Phosphinic acid is a diprotic acid while phosphonic acid is a monoprotic acid.
 (C) Both are diprotic acids.
 (D) Both are triprotic acids.

73. Which of the following reagents would distinguish cis-cyclopenta-1,2-diol from the trans-isomer?

- (A) MnO_2
 (B) Aluminium isopropoxide
 (C) Acetone
 (D) Ozone



74. Which one of the following orders is CORRECT for the bond dissociation enthalpy of halogen molecules?

- (A) $\text{Br}_2 > \text{I}_2 > \text{F}_2 > \text{Cl}_2$
 (B) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$
 (C) $\text{I}_2 > \text{Br}_2 > \text{Cl}_2 > \text{F}_2$
 (D) $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$

75. Which of the following is an analgesic?

- (A) Streptomycin (B) Chloromycetin
 (C) Novalgin (D) Penicillin

76. Consider the following liquid-vapour equilibrium.

Liquid $\rightleftharpoons$ Vapour

Which of the following relation is correct?

- (A) $\frac{d \ln P}{dT} = \frac{-\Delta H_V}{T^2}$ (B) $\frac{d \ln P}{dT} = \frac{\Delta H_V}{RT^2}$
 (C) $\frac{d \ln G}{dT} = \frac{\Delta H_V}{RT^2}$ (D) $\frac{d \ln P}{dT} = \frac{-\Delta H_V}{RT}$

77. Match items of Column I with the items of Column II and assign the correct code:

	Column I		Column II
i.	Cyanide process	a.	Ultrapure Ge
ii.	Froth floatation process	b.	Dressing of ZnS
iii.	Electrolytic reduction	c.	Extraction of Al
iv.	Zone refining	d.	Extraction of Au
		e.	Purification of Ni

- (A) i - a, ii - b, iii - c, iv - d
 (B) i - c, ii - d, iii - e, iv - a
 (C) i - d, ii - b, iii - c, iv - a
 (D) i - b, ii - c, iii - a, iv - e

78. Choose the CORRECT statement regarding a carbonyl compound with a hydrogen atom on its alpha-carbon atom.

- (A) A carbonyl compound with a hydrogen atom on its alpha-carbon rapidly equilibrates with its corresponding enol and this process is known as carbonylation.
 (B) A carbonyl compound with a hydrogen atom on its alpha-carbon rapidly equilibrates with its corresponding enol and this process is known as keto-enol tautomerism.
 (C) A carbonyl compound with a hydrogen atom on its alpha-carbon never equilibrates with its corresponding enol.
 (D) A carbonyl compound with a hydrogen atom on its alpha-carbon rapidly equilibrates with its corresponding enol and this process is known as aldehyde-ketone equilibration.

79. The CORRECT statement regarding RNA and DNA, is _____.

- (A) the sugar component in RNA is arabinose and the sugar component in DNA is ribose

(B) the sugar component in RNA is 2'-deoxyribose and the sugar component in DNA is arabinose

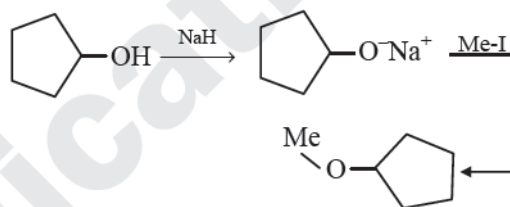
(C) the sugar component in RNA is arabinose and the sugar component in DNA is 2'-deoxyribose

(D) the sugar component in RNA is ribose and the sugar component in DNA is 2'-deoxyribose

80. Lithium has a bcc structure. Its density is 530 kg m^{-3} and its atomic mass is 6.94 g mol^{-1} . Calculate the edge length of a unit cell of lithium metal. ($N_0 = 6.02 \times 10^{23} \text{ mol}^{-1}$)

- (A) 527 pm (B) 264 pm
 (C) 154 pm (D) 352 pm

81. The following reaction can be classified as _____.



- (A) dehydration reaction
 (B) Williamson alcohol synthesis reaction
 (C) Williamson ether synthesis reaction
 (D) alcohol formation reaction

82. At 100°C , the vapour pressure of a solution of 6.5 g of a solute in 100 g water is 732 mm. If $K_b = 0.52$, the boiling point of this solution will be _____.

- (A) 102°C (B) 103°C
 (C) 101°C (D) 100°C

83. Among the following, the CORRECT order of acidity is _____.

- (A) $\text{HClO}_2 < \text{HClO} < \text{HClO}_3 < \text{HClO}_4$
 (B) $\text{HClO}_4 < \text{HClO}_2 < \text{HClO} < \text{HClO}_3$
 (C) $\text{HClO}_3 < \text{HClO}_4 < \text{HClO}_2 < \text{HClO}$
 (D) $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$

84. Natural rubber has _____.

- (A) alternate cis- and trans-configuration
 (B) random cis- and trans-configuration
 (C) all cis-configuration
 (D) all trans-configuration

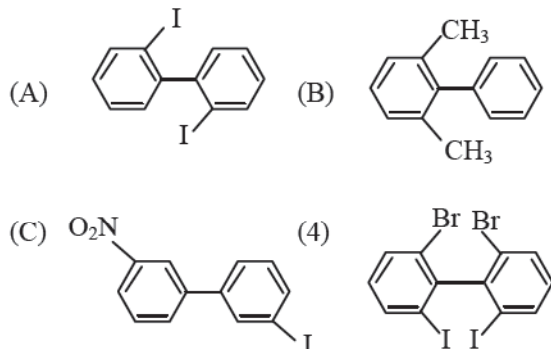
85. Consider the nitration of benzene using mixed conc. H_2SO_4 and HNO_3 . If a large amount of KHSO_4 is added to the mixture, the rate of nitration will be _____.

- (A) unchanged (B) doubled
 (C) faster (D) slower



86. Which one of the following characteristics is associated with adsorption?
- (A) ΔG and ΔH are negative, ΔS is positive.
 (B) ΔG and ΔS are negative, ΔH is positive.
 (C) ΔG is negative but ΔH and ΔS are positive.
 (D) ΔG , ΔH and ΔS all are negative.

87. Which of the following biphenyls is optically active?



88. The ionic radii of A^+ and B^- ions are 0.98×10^{-10} m and 1.81×10^{-10} m. The coordination number of each ion in AB is _____.

(A) 8 (B) 2 (C) 6 (D) 4

89. Which one of the following statements is CORRECT when SO_2 is passed through acidified $K_2Cr_2O_7$ solution?
- (A) SO_2 is reduced.
 (B) Green $Cr_2(SO_4)_3$ is formed.
 (C) The solution turns blue.
 (D) The solution is decolourized.

90. Which of the following statements about hydrogen is INCORRECT?
- (A) Hydrogen has three isotopes of which tritium is the most common.
 (B) Hydrogen never acts as cation in ionic salts.
 (C) Hydronium ion, H_3O^+ exists freely in solution.
 (D) Dihydrogen does not act as a reducing agent.

91. Nomenclature is governed by certain universal rules. Which one of the following is contrary to the rules of nomenclature?
- (A) The names are written in Latin and are italicized.
 (B) When written by hand, the names are to be underlined.
 (C) Biological names can be written in any language.
 (D) The first word in a biological name, represents the genus name and the second is a specific epithet.

92. Which of the following statements is NOT correct?

- (A) Pollen germination and pollen tube growth are regulated by chemical components of pollen interacting with those of the pistil.
 (B) Some reptiles have also been reported as pollinators in some plant species.
 (C) Pollen grains of many species can germinate on the stigma of a flower, but only one pollen tube of the same species grows into the style.
 (D) Insects that consume pollen or nectar without bringing about pollination are called pollen/nectar robbers.

93. The primitive prokaryotes responsible for the production of biogas from the dung of ruminant animals, include the

- (A) Methanogens
 (B) Eubacteria
 (C) Halophiles
 (D) Thermoacidophiles

94. Which of the following approaches does not give the defined action of contraceptive?

(1)	Hormonal contraceptives	Prevent/retard entry of sperms, prevent ovulation and fertilization
(2)	Vasectomy	Prevents spermatogenesis
(3)	Barrier methods	Prevent fertilization
(4)	Intra Uterine Devices	Increase phagocytosis of sperms, suppress sperm motility and fertilizing capacity of sperms

95. Which one of the following characteristics is NOT shared by birds and mammals?

- (A) Viviparity
 (B) Warm blooded nature
 (C) Ossified endoskeleton
 (D) Breathing using lungs

96. Which of the following statements is WRONG for viroids?

- (A) They cause infections.
 (B) Their RNA is of high molecular weight.
 (C) They lack a protein coat.
 (D) They are smaller than viruses.

97. In bryophytes and pteridophytes, transport of male gametes requires

- (A) birds (B) water
 (C) wind (D) insects

98. A complex of ribosomes attached to a single strand of RNA is known as

- (A) polypeptide
 (B) Okazaki fragment
 (C) polysome
 (D) polymer



99. Which type of tissue correctly matches with its location?

	Tissue	Location
(1)	Transitional epithelium	Tip of nose
(2)	Cuboidal epithelium	Lining of stomach
(3)	Smooth muscles	Wall of intestine
(4)	Areolar tissue	Tendons

100. Analogous structures are a result of
(A) Shared ancestry
(B) Stabilizing selection
(C) Divergent evolution
(D) Convergent evolution
101. Which one of the following cell organelles is enclosed by a single membrane?
(A) Lysosomes (B) Nuclei
(C) Mitochondria (D) Chloroplasts
102. A system of rotating crops with legume or grass pasture to improve soil structure and fertility is called:
(A) Strip farming
(B) Shifting agriculture
(C) Ley farming
(D) Contour farming
103. A typical fat molecule is made up of
(A) One glycerol and one fatty acid molecule
(B) Three glycerol and three fatty acid molecules
(C) Three glycerol molecules and one fatty acid molecule
(D) One glycerol and three fatty acid molecules
104. Antivenom injection contains preformed antibodies while polio drops that are administered into the body contain
(A) gamma globulin
(B) attenuated pathogens
(C) activated pathogens
(D) harvested antibodies
105. Which of the following is NOT a characteristic feature during mitosis in somatic cells?
(A) Chromosome movement
(B) Synapsis
(C) Spindle fibres
(D) Disappearance of nucleolus
106. Which of the following is wrongly matched in the given table?

	Microbe	Product	Application
(1)	<i>Streptococcus</i>	Streptokinase	Removal of clot from blood vessel
(2)	<i>Clostridium butylicum</i>	Lipase	Removal of oil stains

(3)	<i>Trichoderma polysporum</i>	Cyclosporin A	Immunosuppressive drug
(4)	<i>Monascus purpureus</i>	Statins	Lowering of blood cholesterol

107. In which of the following, all three are macronutrients?
(A) Boron, zinc, manganese
(B) Iron, copper, molybdenum
(C) Molybdenum, magnesium, manganese
(D) Nitrogen, nickel, phosphorus
108. Which part of the tobacco plant is infected by *Meloidogyne incognita*?
(A) Stem (B) Root
(C) Flower (D) Leaf
109. The *Avena* curvature is used for bioassay of
(A) IAA (B) ethylene
(C) ABA (D) GA₃
110. The term ecosystem was coined by:
(A) E. Haeckel (B) E. Warming
(C) E.P. Odum (D) A.G. Tansley
111. Which of the following guards the opening of hepatopancreatic duct into the duodenum?
(A) Pyloric sphincter
(B) Sphincter of Oddi
(C) Semilunar valve
(D) Ileocaecal valve
112. Which of the following is the most important cause of animals and plants being driven to extinction?
(A) Habitat loss and fragmentation
(B) Co-extinction
(C) Over-exploitation
(D) Alien species invasion
113. Reduction in pH of blood will
(A) decrease the affinity of hemoglobin with oxygen.
(B) release bicarbonate ions by the liver.
(C) reduce the rate of heart beat.
(D) reduce the blood supply to the brain.
114. Which one of the following statements is NOT true?
(A) Pollen grains of many species cause severe allergies.
(B) Stored pollen in liquid nitrogen can be used in the crop breeding programmes.
(C) Tapetum helps in the dehiscence of anther.
(D) Exine of pollen grains is made up of sporopollenin.



115. Which of the following features is NOT present in the phylum Arthropoda?
 (A) Parapodia
 (B) Jointed appendages
 (C) Chitinous exoskeleton
 (D) Metameric segmentation
116. Select the INCORRECT statement:
 (A) LH and FSH decrease gradually during the follicular phase.
 (B) LH triggers secretion of androgens from the Leydig cells.
 (C) FSH stimulates the Sertoli cells which help in spermiogenesis.
 (D) LH triggers ovulation in ovary.
117. In mammals, which blood vessel would normally carry largest amount of urea?
 (A) Hepatic Vein
 (B) Hepatic Portal Vein
 (C) Renal Vein
 (D) Dorsal Aorta
118. Match the term in Column I with their description in Column II and choose the correct option.

	Column I		Column II
a.	Dominance	i.	Many genes govern a single character
b.	Co-dominance	ii.	In a heterozygous organism only one allele expresses itself
c.	Pleiotropy	iii.	In a heterozygous organism both alleles express themselves fully
d.	Polygenic inheritance	iv.	A single gene influences many characters

- (A) a – iv, b – i, c – ii, d – iii
 (B) a – iv, b – iii, c – i, d – ii
 (C) a – ii, b – i, c – iv, d – iii
 (D) a – ii, b – iii, c – iv, d – i
119. Lack of relaxation between successive stimuli sustained muscle contraction is known as
 (A) tetanus (B) tonus
 (C) spasm (D) fatigue
120. In higher vertebrates, the immune system can distinguish self-cells and non-self. If this property is lost due to genetic abnormality and it attacks self-cells, then it leads to
 (A) auto-immune disease
 (B) active immunity
 (C) allergic response
 (D) graft rejection
121. Photosensitive compound in human eye is made up of
 (A) opsin and retinol
 (B) transducin and retinene

- (C) guanosine and retinol
 (D) opsin and retinal
122. Which of the following is NOT a feature of the plasmids?
 (A) Transferable
 (B) Single-stranded
 (C) Independent replication
 (D) Circular structure
123. Identify the CORRECT statement on inhibin.
 (A) Is produced by granulosa cells in ovary and inhibits the secretion of LH.
 (B) Is produced by nurse cells in testes and inhibits the secretion of LH.
 (C) Inhibits the secretion of LH, FSH and Prolactin.
 (D) Is produced by granulosa cells in ovary and inhibits the secretion of FSH.
124. The two polypeptides of human insulin are linked together by
 (A) covalent bond
 (B) disulphide bridges
 (C) hydrogen bonds
 (D) phosphodiester bond
125. Which of the following pairs of hormones are NOT antagonistic (having opposite effects) to each other?
 (A) Aldosterone – Atrial Natriuretic Factor
 (B) Relaxin – Inhibin
 (C) Parathormone – Calcitonin
 (D) Insulin – Glucagon
126. Chrysophytes, Euglenoids, Dinoflagellates and Slime moulds are included in the kingdom
 (A) Fungi (B) Animalia
 (C) Monera (D) Protista
127. It is much easier for a small animal to run uphill than for a large animal, because
 (A) small animals have a lower O_2 requirement.
 (B) the efficiency of muscles in large animals is less than in the small animals.
 (C) it is easier to carry a small body weight.
 (D) smaller animals have a higher metabolic rate.
128. Select the CORRECT statement.
 (A) *Sequoia* is one of the tallest trees.
 (B) The leaves of gymnosperms are not well adapted to extremes of climate.
 (C) Gymnosperms are both homosporous and heterosporous.
 (D) *Salvinia*, *Ginkgo* and *Pinus* all are gymnosperms.



129. Which one of the following is a characteristic feature of cropland ecosystem?
(A) Absence of weeds
(B) Ecological succession
(C) Absence of soil organisms
(D) Least genetic diversity
130. Which is the National Aquatic Animal of India?
(A) Gangetic shark (B) River dolphin
(C) Blue whale (D) Sea – horse
131. A river with an inflow of domestic sewage rich in organic waste may result in:
(A) An increased production of fish due to biodegradable nutrients.
(B) Death of fish due to lack of oxygen.
(C) Drying of the river very soon due to algal bloom.
(D) Increased population of aquatic food web organisms.
132. Stems modified into flat green organs performing the functions of leaves are known as
(A) phylloclades (B) scales
(C) cladodes (D) phyllodes
133. Proximal end of the filament of stamen is attached to the
(A) placenta
(B) thalamus or petal
(C) anther
(D) connective
134. Specialised epidermal cells surrounding the guard cells are called
(A) Bulliform cells
(B) Lenticels
(C) Complementary cells
(D) Subsidiary cells
135. Fertilization in humans is practically feasible only if:
(A) The sperms are transported into vagina just after the release of ovum in fallopian tube
(B) The ovum and sperms are transported simultaneously to ampullary – isthmic junction of the fallopian tube
(C) The ovum and sperms are transported simultaneously to ampullary – isthmic junction of the cervix
(D) The sperms are transported into cervix within 48 hrs. of release of ovum in uterus
136. Changes in GnRH pulse frequency in females is controlled by circulating levels of
(A) progesterone only
(B) progesterone and inhibin
(C) estrogen and progesterone
(D) estrogen and inhibin.
137. Which of the following most appropriately describes haemophilia?
(A) Chromosomal disorder
(B) Dominant gene disorder
(C) Recessive gene disorder
(D) X- linked recessive gene disorder
138. The amino acid tryptophan is the precursor for the synthesis of
(A) estrogen and progesterone
(B) cortisol and cortisone
(C) melatonin and serotonin
(D) thyroxine and triiodothyronine
139. Which one of the following is the starter codon?
(A) UAA (B) UAG
(C) AUG (D) UGA
140. In meiosis, crossing over is initiated at
(A) zygotene (B) diplotene
(C) pachytene (D) leptotene
141. Which of the following structures is homologous to the wing of a bird?
(A) Hind limb of Rabbit
(B) Flipper of Whale
(C) Dorsal fin of a Shark
(D) Wing of a Moth
142. The standard petal of a papilionaceous corolla is also called
(A) vexillum (B) corona
(C) carina (D) pappus
143. The coconut water from tender coconut represents
(A) free nuclear proembryo
(B) free nuclear endosperm
(C) endocarp
(D) fleshy mesocarp
144. Asthma may be attributed to
(A) inflammation of the trachea
(B) accumulation of fluid in the lungs
(C) bacterial infection of the lungs
(D) allergic reaction of the mast cells in the lungs
145. A tall true breeding garden pea plant is crossed with a dwarf true breeding garden pea plant. When the F_1 plants were selfed the resulting genotypes were in the ratio of
(A) 3 : 1 :: Tall : Dwarf
(B) 3 : 1 :: Dwarf : Tall
(C) 1 : 2 : 1 :: Tall homozygous : Tall heterozygous : Dwarf
(D) 1 : 2 : 1 :: Tall heterozygous : Tall homozygous : Dwarf



146. Blood pressure in the pulmonary artery is
(A) more than that in pulmonary vein
(B) less than that in vena cavae
(C) same as that in the aorta
(D) more than that in the carotid
147. Which of the following is required as inducer(s) for the expression of *Lac* operon?
(A) Lactose
(B) Lactose and Galactose
(C) Glucose
(D) Galactose
148. Microtubules are the constituents of
(A) Centrioles, spindle fibres and chromatin
(B) Centrosome, nucleosome and centrioles
(C) Cilia, flagella and peroxisomes
(D) Spindle fibres, centrioles and cilia
149. In context of amniocentesis, which of the following statement is INCORRECT?
(A) It can be used for detection of Down syndrome.
(B) It can be used for detection of Cleft palate.
(C) It is usually done when a woman is between 14 - 16 weeks pregnant.
(D) It is used for prenatal sex determination.
150. Which one of the following statements is WRONG?
(A) Uracil is a pyrimidine.
(B) Glycine is a sulphur containing amino acid.
(C) Sucrose is a disaccharide.
(D) Cellulose is a polysaccharide.
151. In a chloroplast, the highest number of protons are found in
(A) inter membrane space
(B) antennae complex
(C) stroma
(D) lumen of thylakoids
152. In a test cross involving F_1 dihybrid flies, more parental-type offspring were produced than the recombinant-type offspring. This indicates
(A) The two genes are linked and present on the same chromosome.
(B) Both the characters are controlled by more than one gene.
(C) The two genes are located on two different chromosomes.
(D) Chromosomes failed to separate during meiosis.
153. In stomach, gastric acid is secreted by
(A) peptic cells
(B) acidic cells
(C) gastric secreting cells
(D) parietal cells
154. Which of the following is not required for any of the techniques of DNA fingerprinting available at present?
(A) Restriction enzymes
(B) DNA – DNA hybridisation
(C) Polymerase chain reaction
(D) Zinc finger analysis
155. Water vapour comes out from the plant leaf through the stomatal opening. Through the same stomatal opening carbon dioxide diffuses into the plant during photosynthesis. Reason out the above statements using one of following options:
(A) The above processes happen only during night time.
(B) One process occurs during day time, and the other at night.
(C) Both processes cannot happen simultaneously.
(D) Both processes can happen together because the diffusion coefficient of water and CO_2 is different.
156. The *Taq* polymerase enzyme is obtained from
(A) *Bacillus subtilis*
(B) *Pseudomonas putida*
(C) *Thermus aquaticus*
(D) *Thiobacillus ferrooxidans*
157. A plant in your garden avoids photorespiratory losses, has improved water use efficiency, shows high rates of photosynthesis at high temperatures and has improved efficiency of nitrogen utilisation. In which of the following physiological groups would you assign this plant?
(A) CAM (B) Nitrogen fixer
(C) C_3 (D) C_4
158. Which of the following feature is NOT present in *Periplaneta americana*?
(A) Exoskeleton composed at N-acetyl glucosamine.
(B) Metamerically segmented body.
(C) Schizocoelom as a body cavity.
(D) Indeterminate and radial cleavage during embryonic development.
159. Pick out the correct statements:
a. Haemophilia is a sex-linked recessive disease.
b. Down's syndrome is due to aneuploidy.
c. Phenylketonuria is an autosomal recessive gene disorder.
d. Sickle cell anaemia is an X- linked recessive gene disorder.
(A) a, c and d are correct.
(B) a, b and c are correct.
(C) a and d are correct.
(D) b and d are correct.



160. Which of the following is NOT a stem modification?
 (A) Tendrils of cucumber
 (B) Flattened structures of *Opuntia*
 (C) Pitcher of *Nepenthes*
 (D) Thorns of *Citrus*
161. When does the growth rate of a population following the logistic model equal zero? The logistic model is given as $dN/dt = rN(1-N/K)$
 (A) when N/K equals zero
 (B) when death rate is greater than birth rate
 (C) when N/K is exactly one
 (D) when N nears the carrying capacity of the habitat
162. Spindle fibres attach on to
 (A) centromere of the chromosome
 (B) kinetosome of the chromosome
 (C) telomere of the chromosome
 (D) kinetochore of the chromosome
163. Which of the following is a restriction endonuclease?
 (A) DNase I (B) RNase
 (C) *Hind* II (D) Protease
164. Water soluble pigments found in plant cell vacuoles are:
 (A) carotenoids (B) anthocyanins
 (C) xanthophylls (D) chlorophylls
165. Depletion of which gas in the atmosphere can lead to an increased incidence of skin cancers?
 (A) Ammonia (B) Methane
 (C) Nitrous oxide (D) Ozone
166. Which one of the following statements is wrong?
 (A) Eubacteria are also called false bacteria.
 (B) Phycomycetes are also called algal fungi.
 (C) Cyanobacteria are also called blue-green algae.
 (D) Golden algae are also called desmids.
167. Seed formation without fertilization in flowering plants involves the process of
 (A) Somatic hybridization
 (B) Apomixis
 (C) Sporulation
 (D) Budding
168. Tricarpellary, syncarpous gynoecium is found in flowers of
 (A) Fabaceae (B) Poaceae
 (C) Liliaceae (D) Solanaceae
169. Emerson's enhancement effect and Red drop have been instrumental in the discovery of:
 (A) photophosphorylation and cyclic electron transport
 (B) oxidative phosphorylation
 (C) photophosphorylation and non-cyclic electron transport
 (D) two photosystems operating simultaneously
170. Following are the two statements regarding the origin of life:
 (i) The earliest organisms that appeared on the earth were non-green and presumably anaerobes.
 (ii) The first autotrophic organisms were the chemoautotrophs that never released oxygen.
 Of the above statements which one of the following options is correct?
 (A) Both (i) and (ii) are correct.
 (B) Both (i) and (ii) are false.
 (C) (i) is correct but (ii) is false.
 (D) (ii) is correct but (i) is false.
171. One of the major components of cell wall of most fungi is
 (A) chitin (B) peptidoglycan
 (C) cellulose (D) hemicellulose
172. Which of the following statements is NOT true for cancer cells in relation to mutations?
 (A) Mutations inactivate the cell control
 (B) Mutations inhibit production of telomerase
 (C) Mutations in proto-oncogenes accelerate the cell cycle.
 (D) Mutations destroy telomerase inhibitor.
173. Name the chronic respiratory disorder caused mainly by cigarette smoking.
 (A) Respiratory acidosis
 (B) Respiratory alkalosis
 (C) Emphysema
 (D) Asthma
174. Which of the following characteristic features always holds TRUE for the corresponding group of animals?
- | | | |
|-----|---|----------------|
| (1) | Possess a mouth with an upper and a lower jaw | Chordata |
| (2) | 3-chambered heart with One incompletely divided ventricle | Reptilia |
| (3) | Cartilaginous Endoskeleton | Chondrichthyes |
| (4) | Viviparous | Mammalia |



175. Gause's principle of competitive exclusion states that
- (A) no two species can occupy the same niche indefinitely for the same limiting resources.
 - (B) larger organisms exclude smaller ones through competition.
 - (C) more abundant species will exclude the less abundant species through competition.
 - (D) competition for the same resources excludes species having different food preferences.
176. Cotyledon of maize grain is called
- (A) coleoptile
 - (B) scutellum
 - (C) plumule
 - (D) coleorhiza
177. A cell at telophase stage is observed by a student in a plant brought from the field. He tells his teacher that this cell is not like other cells at telophase stage. There is no formation of cell plate and thus the cell is containing one number of chromosomes as compared to other dividing cells. This would result in
- (A) Somaclonal variation
 - (B) Polyteny
 - (C) Aneuploidy
 - (D) Polyploidy
178. Mitochondria and chloroplast are
- i. semi-autonomous organelles
 - ii. formed by division of pre-existing organelles and they contain DNA but lack protein synthesizing machinery.
- Which one of the following options is correct?
- (A) (i) is true but (ii) is false
 - (B) Both (i) and (ii) are false
 - (C) Both (i) and (ii) are correct
 - (D) (ii) is true but (i) is false
179. Which of the following would appear as the pioneer organisms on bare rocks?
- (A) Mosses
 - (B) Green algae
 - (C) Lichens
 - (D) Liverworts
180. Joint Forest Management Concept was introduced in India during:
- (A) 1980s
 - (B) 1990s
 - (C) 1960s
 - (D) 1970s

NEET (UG) – 2016 (Phase II) Question Paper

1. Photons with energy 5 eV are incident on a cathode C in a photoelectric cell. The maximum energy of emitted photoelectrons is 2 eV. When photons of energy 6 eV are incident on C, no photoelectrons will reach the anode A, if the stopping potential of A relative to C is
(A) -3 V (B) +3 V (C) +4 V (D) -1 V

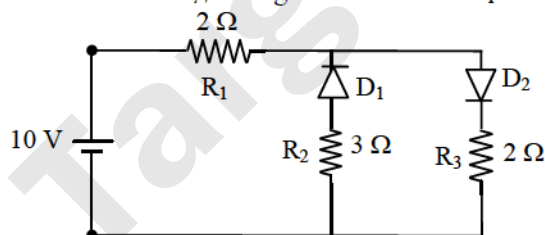
2. Three sound waves of equal amplitudes have frequencies $(n-1)$, n , $(n+1)$. They superimpose to give beats. The number of beats produced per second will be
(A) 2 (B) 1 (C) 4 (D) 3

3. A person can see clearly objects only when they lie between 50 cm and 400 cm from his eyes. In order to increase the maximum distance of distinct vision to infinity, the type and power of the correcting lens, the person has to use, will be
(A) convex, +0.15 dioptre
(B) convex, +2.25 dioptre
(C) concave, -0.25 dioptre
(D) concave, -0.2 dioptre

4. Two cars P and Q start from a point at the same time in a straight line and their positions are represented by $x_P(t) = at + bt^2$ and $x_Q(t) = ft - t^2$. At what time do the cars have the same velocity?

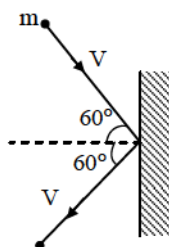
- (A) $\frac{f-a}{2(1+b)}$ (B) $\frac{a-f}{1+b}$
(C) $\frac{a+f}{2(b-1)}$ (D) $\frac{a+f}{2(1+b)}$

5. The given circuit has two ideal diodes connected as shown in the figure below. The current flowing through the resistance R_1 will be



- (A) 3.13 A (B) 2.5 A
(C) 10.0 A (D) 1.43 A

6. A rigid ball of mass m strikes a rigid wall at 60° and gets reflected without loss of speed as shown in the figure below. The value of impulse imparted by the wall on the ball will be

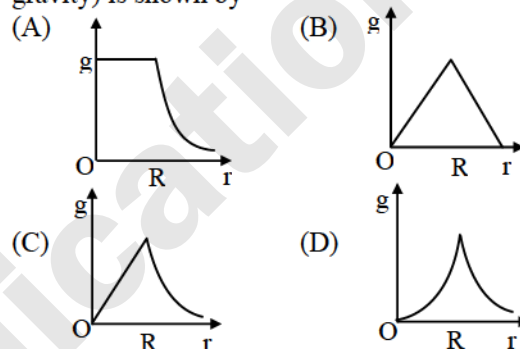


- (A) $\frac{mV}{3}$ (B) mV (C) $2mV$ (D) $\frac{mV}{2}$

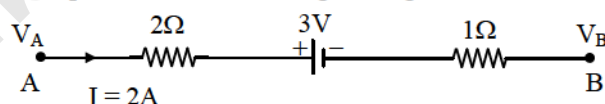
7. An electric dipole is placed at an angle of 30° with an electric field intensity 2×10^5 N/C. It experiences a torque equal to 4 Nm. The charge on the dipole, if the dipole length is 2 cm, is

- (A) $7 \mu\text{C}$ (B) 8 mC
(C) 2 mC (D) 5 mC

8. Starting from the centre of the earth having radius R , the variation of g (acceleration due to gravity) is shown by



9. The potential difference ($V_A - V_B$) between the points A and B in the given figure is



- (A) +9 V (B) -3 V
(C) +3 V (D) +6 V

10. A rectangular film of liquid is extended from $(4 \text{ cm} \times 2 \text{ cm})$ to $(5 \text{ cm} \times 4 \text{ cm})$. If the work done is 3×10^{-4} J, the value of the surface tension of the liquid is

- (A) 8.0 N m^{-1} (B) 0.250 N m^{-1}
(C) 0.125 N m^{-1} (D) 0.2 N m^{-1}

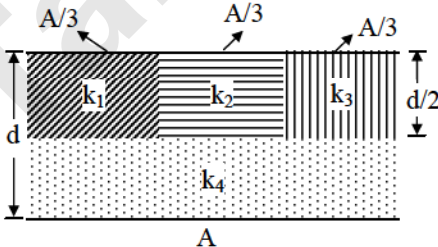
11. For CE transistor amplifier, the audio signal voltage across the collector resistance of $2 \text{ k}\Omega$ is 4 V. If the current amplification factor of the transistor is 100 and the base resistance is $1 \text{ k}\Omega$, then the input signal voltage is

- (A) 15 mV (B) 10 mV
(C) 20 mV (D) 30 mV

12. A light rod of length l has two masses m_1 and m_2 attached to its two ends. The moment of inertia of the system about an axis perpendicular to the rod and passing through the centre of mass is

- (A) $\sqrt{m_1 m_2} l^2$ (B) $\frac{m_1 m_2}{m_1 + m_2} l^2$
(C) $\frac{m_1 + m_2}{m_1 m_2} l^2$ (D) $(m_1 + m_2) l^2$



13. One mole of an ideal monatomic gas undergoes a process described by the equation $PV^3 = \text{constant}$. The heat capacity of the gas during this process is
 (A) R (B) $\frac{3}{2}R$
 (C) $\frac{5}{2}R$ (D) $2R$
14. A linear aperture whose width is 0.02 cm is placed immediately in front of a lens of focal length 60 cm. The aperture is illuminated normally by a parallel beam of wavelength 5×10^{-5} cm. The distance of the first dark band of the diffraction pattern from the centre of the screen is
 (A) 0.15 cm (B) 0.10 cm
 (C) 0.25 cm (D) 0.20 cm
15. The second overtone of an open organ pipe has the same frequency as the first overtone of a closed pipe L metre long. The length of the open pipe will be
 (A) $4L$ (B) L
 (C) $2L$ (D) $\frac{L}{2}$
16. A bullet of mass 10 g moving horizontally with a velocity of 400 ms^{-1} strikes a wooden block of mass 2 kg which is suspended by a light inextensible string of length 5 m. As a result, the centre of gravity of the block found to rise a vertical distance of 10 cm. The speed of the bullet after it emerges out horizontally from the block will be
 (A) 160 m s^{-1} (B) 100 m s^{-1}
 (C) 80 m s^{-1} (D) 120 m s^{-1}
17. A parallel - plate capacitor of area A , plate separation d and capacitance C is filled with four dielectric materials having dielectric constants k_1, k_2, k_3 and k_4 as shown in the figure. If a single dielectric material is to be used to have the same capacitance C in this capacitor, then its dielectric constant k is given by
- 
- (A) $k = k_1 + k_2 + k_3 + 3k_4$
 (B) $k = \frac{2}{3}(k_1 + k_2 + k_3) + 2k_4$
 (C) $\frac{2}{k} = \frac{3}{k_1 + k_2 + k_3} + \frac{1}{k_4}$
 (D) $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} + \frac{3}{2k_4}$
18. A satellite of mass m is orbiting the earth (of radius R) at a height h from its surface. the total energy of the satellite in terms of g_0 , the value of acceleration due to gravity at the earth's surface, is
 (A) $-\frac{2mg_0R^2}{R+h}$ (B) $\frac{mg_0R^2}{2(R+h)}$
 (C) $-\frac{mg_0R^2}{2(R+h)}$ (D) $\frac{2mg_0R^2}{R+h}$
19. A filament bulb (500W, 100V) is to be used in a 230 V main supply. When a resistance is connected in series, it works perfectly and the bulb consumes 500 W. The value of R is
 (A) 13Ω (B) 230Ω
 (C) 46Ω (D) 26Ω
20. A solid sphere of mass m and radius R is rotating about its diameter. A solid cylinder of the same mass and same radius is also rotating about its geometrical axis with an angular speed twice that of the sphere. The ratio of their kinetic energies of rotation ($E_{\text{sphere}}/E_{\text{cylinder}}$) will be
 (A) 3:1 (B) 2:3 (C) 1:5 (D) 1:4
21. A long wire carrying a steady current is bent into a circular loop of one turn. The magnetic field at the centre of the loop is B . It is then bent into a circular coil of n turns. The magnetic field at the centre of this coil of n turns will be
 (A) $2n^2B$ (B) nB
 (C) n^2B (D) $2nB$
22. Two identical bodies are made of a material for which the heat capacity increases with temperature. One of these is at 100°C , while the other one is at 0°C . If the two bodies are brought into contact, then, assuming no heat loss, the final common temperature is
 (A) 0°C
 (B) 50°C
 (C) more than 50°C
 (D) less than 50°C but greater than 0°C
23. A bar magnet is hung by a thin cotton thread in a uniform horizontal magnetic field and is in equilibrium. The energy required to rotate it by 60° is W . Now the torque required to keep the magnet in this new position is
 (A) $\frac{2W}{\sqrt{3}}$ (B) $\frac{W}{\sqrt{3}}$ (C) $\sqrt{3}W$ (D) $\frac{\sqrt{3}W}{2}$
24. A given sample of an ideal gas occupies a volume V at a pressure P and absolute temperature T . The mass of each molecule of the gas is m . Which of the following gives the density of the gas?
 (A) mkT (B) $P/(kT)$
 (C) $Pm/(kT)$ (D) $P/(kTV)$

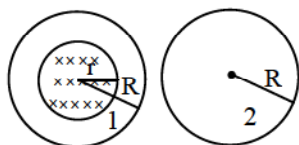
25. If an electron in a hydrogen atom jumps from the 3rd orbit to the 2nd orbit, it emits a photon of wavelength λ . When it jumps from the 4th orbit to the 3rd orbit, the corresponding wavelength of the photon will be

(A) $\frac{20}{13}\lambda$ (B) $\frac{16}{25}\lambda$
(C) $\frac{9}{16}\lambda$ (D) $\frac{20}{7}\lambda$

26. A body cools from a temperature $3T$ to $2T$ in 10 minutes. The room temperature is T . Assume that Newton's law of cooling is applicable. The temperature of the body at the end of next 10 minutes will be

(A) T (B) $\frac{7}{4}T$
(C) $\frac{3}{2}T$ (D) $\frac{4}{3}T$

27. A uniform magnetic field is restricted within a region of radius r . The magnetic field changes with time at a rate $\frac{d\vec{B}}{dt}$. Loop 1 of radius $R > r$ encloses the region r and loop 2 of radius R is outside the region of magnetic field as shown in the figure below. Then the e.m.f. generated is



(A) $-\frac{d\vec{B}}{dt}\pi r^2$ in loop 1 and zero in loop 2
(B) zero in loop 1 and zero in loop 2
(C) $-\frac{d\vec{B}}{dt}\pi r^2$ in loop 1 and $-\frac{d\vec{B}}{dt}\pi R^2$ in loop 2
(D) $-\frac{d\vec{B}}{dt}\pi r^2$ in loop 1 and zero in loop 2

28. Three liquids of densities ρ_1 , ρ_2 and ρ_3 (with $\rho_1 > \rho_2 > \rho_3$), having the same value of surface tension T , rise to the same height in three identical capillaries. The angles of contact θ_1 , θ_2 and θ_3 obey

(A) $\pi > \theta_1 > \theta_2 > \theta_3 > \frac{\pi}{2}$
(B) $\frac{\pi}{2} > \theta_1 > \theta_2 > \theta_3 \geq 0$
(C) $0 \leq \theta_1 < \theta_2 < \theta_3 < \frac{\pi}{2}$
(D) $\frac{\pi}{2} < \theta_1 < \theta_2 < \theta_3 < \pi$

29. The potential differences across the resistance, capacitance and inductance are 80 V, 40 V and 100 V respectively in an L-C-R circuit. The power factor of this circuit is

(A) 1.0 (B) 0.4
(C) 0.5 (D) 0.8

30. Two identical glass ($\mu_g = 3/2$) equiconvex lenses of focal length f each are kept in contact. The space between the two lenses is filled with water ($\mu_w = 4/3$). The focal length of the combination is

(A) $\frac{3f}{4}$ (B) $\frac{f}{3}$ (C) f (D) $\frac{4f}{3}$

31. Two rotating bodies A and B of masses m and $2m$ with moments of inertia I_A and I_B ($I_B > I_A$) have equal kinetic energy of rotation. If L_A and L_B be their angular momenta respectively, then

(A) $L_A > L_B$ (B) $L_A = \frac{L_B}{2}$
(C) $L_A = 2L_B$ (D) $L_B > L_A$

32. The interference pattern is obtained with two coherent light sources of intensity ratio n . In the interference pattern, the ratio $\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$ will be

(A) $\frac{2\sqrt{n}}{(n+1)^2}$ (B) $\frac{\sqrt{n}}{n+1}$
(C) $\frac{2\sqrt{n}}{n+1}$ (D) $\frac{\sqrt{n}}{(n+1)^2}$

33. Two identical balls A and B having velocities of 0.5 m/s and -0.3 m/s respectively collide elastically in one dimension. The velocities of B and A after the collision respectively will be

(A) 0.3 m/s and 0.5 m/s
(B) -0.5 m/s and 0.3 m/s
(C) 0.5 m/s and -0.3 m/s
(D) -0.3 m/s and 0.5 m/s

34. Which of the following combinations should be selected for better tuning of an L-C-R circuit used for communication?

(A) $R = 25 \Omega$, $L = 1.5$ H, $C = 45 \mu\text{F}$
(B) $R = 20 \Omega$, $L = 1.5$ H, $C = 35 \mu\text{F}$
(C) $R = 25 \Omega$, $L = 2.5$ H, $C = 45 \mu\text{F}$
(D) $R = 15 \Omega$, $L = 3.5$ H, $C = 30 \mu\text{F}$

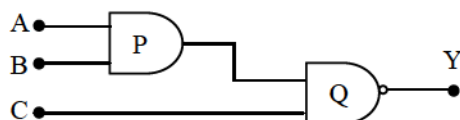
35. The half-life of a radioactive substance is 30 minutes. The time (in minutes) taken between 40% decay and 85% decay of the same radioactive substance is

(A) 60 (B) 15
(C) 30 (D) 45

36. Planck's constant (h), speed of light in vacuum (c) and Newton's gravitational constant (G) are three fundamental constants. Which of the following combinations of these has the dimension of length?

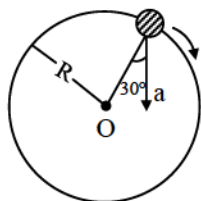
(A) $\sqrt{\frac{Gc}{h^{3/2}}}$ (B) $\frac{\sqrt{hG}}{c^{3/2}}$
(C) $\frac{\sqrt{hG}}{c^{5/2}}$ (D) $\sqrt{\frac{hc}{G}}$

37. What is the output Y in the following circuit, when all the three inputs A, B, C are first 0 and then 1?



- (A) 1, 1 (B) 0, 1
(C) 0, 0 (D) 1, 0

38. In the given figure, $a = 15 \text{ m/s}^2$ represents the total acceleration of a particle moving in the clockwise direction in a circle of radius $R = 2.5 \text{ m}$ at a given instant of time. the speed of the particle is



- (A) 6.2 m/s
(B) 4.5 m/s
(C) 5.0 m/s
(D) 5.7 m/s
39. An electron is moving in a circular path under the influence of a transverse magnetic field of $3.57 \times 10^{-2} \text{ T}$. If the value of e/m is $1.76 \times 10^{11} \text{ C/kg}$, the frequency of revolution of the electron is
- (A) 6.28 MHz (B) 1 GHz
(C) 100 MHz (D) 62.8 MHz

40. A body of mass m is attached to the lower end of a spring whose upper end is fixed. The spring has negligible mass. When the mass m is slightly pulled down and released, it oscillates with a time period of 3 s. When the mass m is increased by 1 kg, the time period of oscillations becomes 5 s. The value of m in kg is

- (A) $\frac{9}{16}$ (B) $\frac{3}{4}$ (C) $\frac{4}{3}$ (D) $\frac{16}{9}$

41. Electrons of mass m with de-Broglie wavelength λ fall on the target in an X-ray tube. The cut-off wavelength (λ_0) of the emitted X-ray is

- (A) $\lambda_0 = \lambda$ (B) $\lambda_0 = \frac{2mc\lambda^2}{h}$
(C) $\lambda_0 = \frac{2h}{mc}$ (D) $\lambda_0 = \frac{2m^2c^2\lambda^3}{h^2}$

42. An air bubble in a glass slab with refractive index 1.5 (near normal incidence) is 5 cm deep when viewed from one surface and 3 cm deep when viewed from the opposite face. The thickness (in cm) of the slab is

- (A) 16 (B) 8 (C) 10 (D) 12

43. A particle moves from a point $(-2\hat{i} + 5\hat{j})$ to $(4\hat{j} + 3\hat{k})$ when a force of $(4\hat{i} + 3\hat{j}) \text{ N}$ is applied. How much work has been done by the force?

- (A) 2 J (B) 8 J (C) 11 J (D) 5 J

44. A 100Ω resistance and a capacitor of 100Ω reactance are connected in series across a 220V source. When the capacitor is 50% charged, the peak value of the displacement current is

- (A) $11\sqrt{2} \text{ A}$ (B) 2.2 A
(C) 11 A (D) 4.4 A

45. The temperature inside a refrigerator is $t_2^\circ\text{C}$ and the room temperature is $t_1^\circ\text{C}$. The amount of heat delivered to the room for each joule of electrical energy consumed ideally will be

- (A) $\frac{t_1 + t_2}{t_1 + 273}$ (B) $\frac{t_1}{t_1 - t_2}$
(C) $\frac{t_1 + 273}{t_1 - t_2}$ (D) $\frac{t_2 + 273}{t_1 - t_2}$

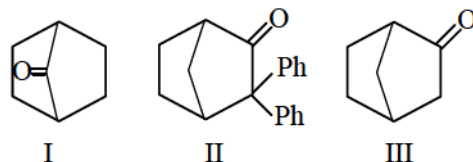
46. Boric acid is an acid because its molecule

- (A) combines with proton from water molecule
(B) contains replaceable H^+ ion
(C) gives up a proton
(D) accepts OH^- from water releasing proton

47. Which of the following can be used as the halide component for Friedel-Crafts reaction?

- (A) Isopropyl chloride
(B) Chlorobenzene
(C) Bromobenzene
(D) Chloroethene

48. Which among the given molecules can exhibit tautomerism?

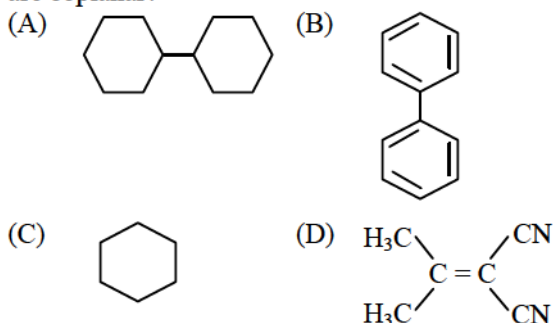


- (A) Both II and III (B) III only
(C) Both I and III (D) Both I and II

49. The types of hybrid orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ respectively are expected to be

- (A) sp , sp^3 and sp^2 (B) sp , sp^2 and sp^3
(C) sp^2 , sp and sp^3 (D) sp^2 , sp^3 and sp

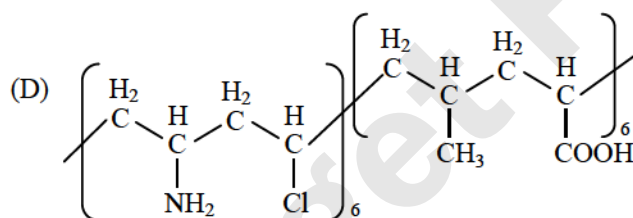
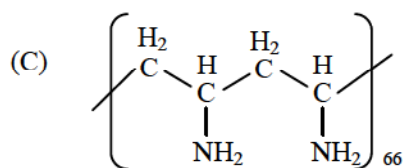
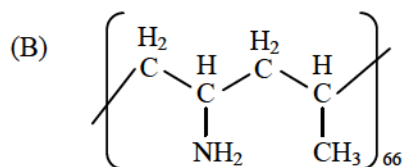
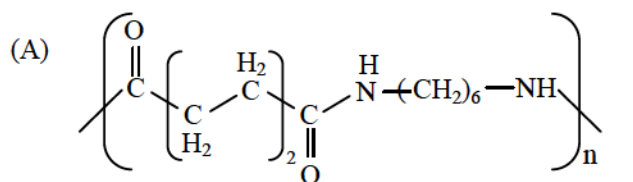
50. In which of the following molecules, all atoms are coplanar?



51. AlF_3 is soluble in HF only in the presence of KF. It is due to the formation of _____.
 (A) $\text{K}[\text{AlF}_3\text{H}]$ (B) $\text{K}_3[\text{AlF}_3\text{H}_3]$
 (C) $\text{K}_3[\text{AlF}_6]$ (D) AlH_3

52. The CORRECT increasing order of trans-effect of the following species is _____.
 (A) $\text{CN}^- > \text{Br}^- > \text{C}_6\text{H}_5^- > \text{NH}_3$
 (B) $\text{NH}_3 > \text{CN}^- > \text{Br}^- > \text{C}_6\text{H}_5^-$
 (C) $\text{CN}^- > \text{C}_6\text{H}_5^- > \text{Br}^- > \text{NH}_3$
 (D) $\text{Br}^- > \text{CN}^- > \text{NH}_3 > \text{C}_6\text{H}_5^-$

53. Which one of the following structures represents nylon-6,6 polymer?



54. Which of the following fluoro compounds is most likely to behave as a Lewis base?
 (A) SiF_4 (B) BF_3
 (C) PF_3 (D) CF_4

55. Zinc can be coated on iron to produce galvanized iron but the reverse is not possible. It is because _____.
 (A) zinc has higher negative electrode potential than iron
 (B) zinc is lighter than iron
 (C) zinc has lower melting point than iron
 (D) zinc has lower negative electrode potential than iron

56. How many electrons can fit in the orbital for which $n = 3$ and $l = 1$?
 (A) 14 (B) 2 (C) 6 (D) 10

57. Which one of the following is INCORRECT for ideal solution?

- (A) $\Delta_{\text{mix}}G = 0$
 (B) $\Delta_{\text{mix}}H = 0$
 (C) $\Delta_{\text{mix}}U = 0$
 (D) $\Delta P = P_{\text{obs}} - P_{\text{calculated by Raoult's law}} = 0$

58. Which one of the following statements related to lanthanoids is INCORRECT?

- (A) Ce (+4) solutions are widely used as oxidizing agent in volumetric analysis.
 (B) Europium shows +2 oxidation state.
 (C) The basicity decreases as the ionic radius decreases from Pr to Lu.
 (D) All the lanthanoids are much more reactive than aluminium.

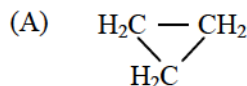
59. The number of electrons delivered at the cathode during electrolysis by a current of 1 ampere in 60 seconds is _____. (charge on electron = 1.60×10^{-19} C)

- (A) 7.48×10^{23} (B) 6×10^{23}
 (C) 6×10^{20} (D) 3.75×10^{20}

60. Suppose the elements X and Y combine to form two compounds XY_2 and X_3Y_2 . When 0.1 mole of XY_2 weighs 10 g and 0.05 mole of X_3Y_2 weighs 9 g, the atomic weights of X and Y are _____.

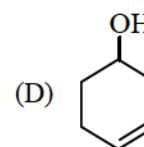
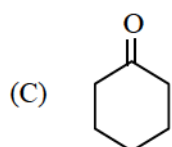
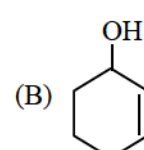
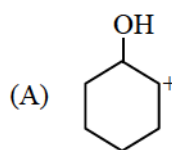
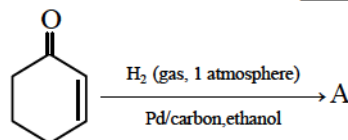
- (A) 30, 20 (B) 40, 30
 (C) 60, 40 (D) 20, 30

61. Which of the following compounds will NOT produce propene by reaction with HBr followed by elimination or direct only elimination reaction?

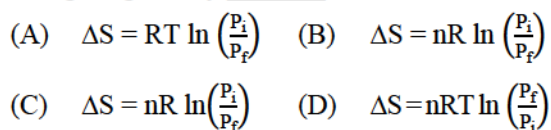


- (B) $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2\text{OH}$
 (C) $\text{H}_2\text{C}=\text{C}=\text{O}$
 (D) $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2\text{Br}$

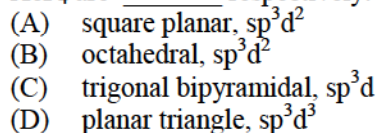
62. The CORRECT structure of the product 'A' formed in the reaction is _____.



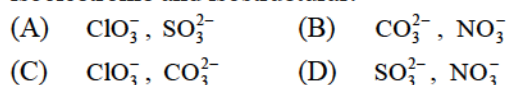
63. For a sample of perfect gas when its pressure is changed isothermally from P_i to P_f , the entropy change is given by _____.



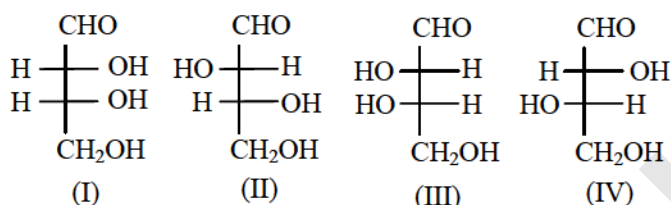
64. The CORRECT geometry and hybridization for XeF_4 are _____ respectively.



65. Which of the following pairs of ions is isoelectronic and isostructural?



66. The CORRECT corresponding order of names of four aldoses with configuration given below:



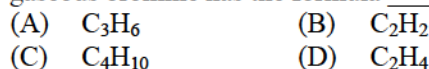
respectively, is _____.

- (A) D-erythrose, D-threose, L-erythrose, L-threose
 (B) L-erythrose, L-threose, L-erythrose, D-threose
 (C) D-threose, D-erythrose, L-threose, L-erythrose
 (D) L-erythrose, L-threose, D-erythrose, D-threose

67. The decomposition of phosphine (PH_3) on tungsten at low pressure is a first order reaction. It is because the _____.

- (A) rate of decomposition is very slow
 (B) rate is proportional to the surface coverage
 (C) rate is inversely proportional to the surface coverage
 (D) rate is independent of the surface coverage

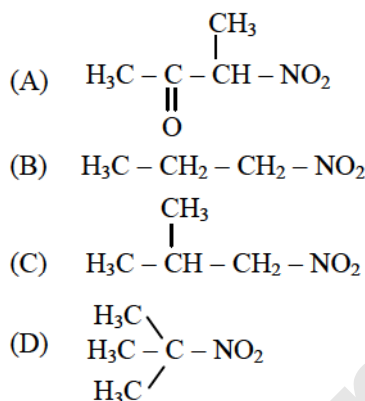
68. The compound that will react most readily with gaseous bromine has the formula _____.



69. If the E°_{cell} for a given reaction has a negative value, which of the following gives the CORRECT relationships for the values of ΔG° and K_{eq} ?

- (A) $\Delta G^\circ < 0$; $K_{\text{eq}} < 1$ (B) $\Delta G^\circ > 0$; $K_{\text{eq}} < 1$
 (C) $\Delta G^\circ > 0$; $K_{\text{eq}} > 1$ (D) $\Delta G^\circ < 0$; $K_{\text{eq}} > 1$

70. Which one of the following nitro compounds does NOT react with nitrous acid?



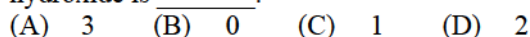
71. Jahn-Teller effect is NOT observed in high spin complexes of _____.



72. In context with beryllium, which one of the following statements is INCORRECT?

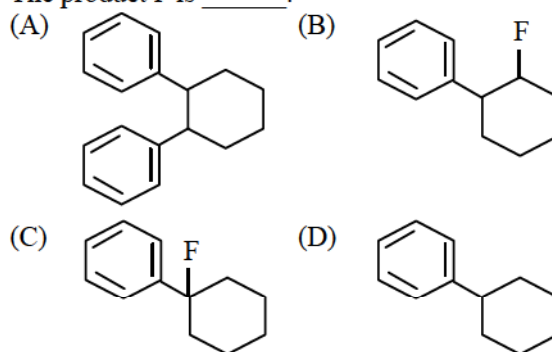
- (A) Its hydride is electron-deficient and polymeric.
 (B) It is rendered passive by nitric acid.
 (C) It forms Be_2C .
 (D) Its salts rarely hydrolyze.

73. The van't Hoff factor (i) for a dilute aqueous solution of the strong electrolyte barium hydroxide is _____.

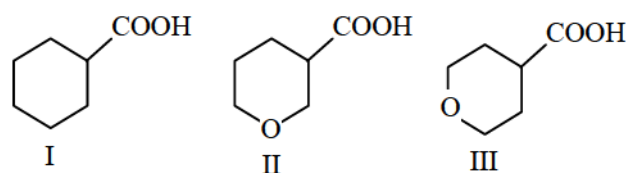


74. $\text{C}_6\text{H}_6 + \text{C}_6\text{H}_{10} \xrightarrow[0^\circ\text{C}]{\text{HF}} \text{P}$

The product P is _____.



75. The CORRECT order of strengths of the carboxylic acids is _____.



- (A) II > I > III (B) I > II > III
 (C) II > III > I (D) III > II > I

76. Hot concentrated sulphuric acid is a moderately strong oxidizing agent. Which of the following reactions does NOT show oxidizing behaviour?

- (A) $\text{CaF}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{CaSO}_4 + 2\text{HF}$
 (B) $\text{Cu} + 2\text{H}_2\text{SO}_4 \longrightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$
 (C) $\text{S} + 2\text{H}_2\text{SO}_4 \longrightarrow 3\text{SO}_2 + 2\text{H}_2\text{O}$
 (D) $\text{C} + 2\text{H}_2\text{SO}_4 \longrightarrow \text{CO}_2 + 2\text{SO}_2 + 2\text{H}_2\text{O}$

77. Which of the following pairs of d-orbitals will have electron density along the axis?

- (A) $d_{xy}, d_{x^2-y^2}$ (B) d_{z^2}, d_{xz}
 (C) d_{xz}, d_{yz} (D) $d_{z^2}, d_{x^2-y^2}$

78. During the electrolysis of molten sodium chloride, the time required to produce 0.10 mol of chlorine gas using a current of 3 amperes is _____.

- (A) 330 minutes (B) 55 minutes
 (C) 110 minutes (D) 220 minutes

79. The central dogma of molecular genetics states that the genetic information flows from _____.

- (A) $\text{DNA} \rightarrow \text{RNA} \rightarrow \text{carbohydrates}$
 (B) $\text{amino acids} \rightarrow \text{proteins} \rightarrow \text{DNA}$
 (C) $\text{DNA} \rightarrow \text{carbohydrates} \rightarrow \text{proteins}$
 (D) $\text{DNA} \rightarrow \text{RNA} \rightarrow \text{proteins}$

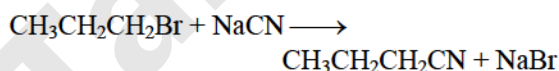
80. Among the following, which one is a INCORRECT statement?

- (A) I_3^+ has bent geometry.
 (B) PH_5 and BiCl_5 do not exist.
 (C) $p\pi-d\pi$ bonds are present in SO_2 .
 (D) SeF_4 and CH_4 have same shape.

81. In calcium fluoride, having the fluorite structure, the coordination numbers for calcium ion (Ca^{2+}) and fluoride ion (F^-) are _____ respectively.

- (A) 4 and 8 (B) 4 and 2
 (C) 6 and 6 (D) 8 and 4

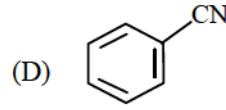
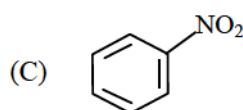
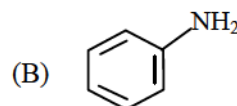
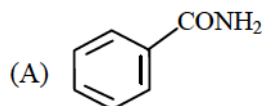
82. Consider the reaction:



This reaction will be the fastest in _____.

- (A) water
 (B) ethanol
 (C) methanol
 (D) N,N'-dimethylformamide (DMF)

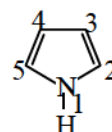
83. A given nitrogen-containing aromatic compound A reacts with Sn/HCl , followed by HNO_2 to give an unstable compound B. B, on treatment with phenol, forms a beautiful coloured compound C with the molecular formula $\text{C}_{12}\text{H}_{10}\text{N}_2\text{O}$. The structure of compound A is _____.



84. Which one of the following compounds shows the presence of intramolecular hydrogen bond?

- (A) Concentrated acetic acid
 (B) H_2O_2
 (C) HCN
 (D) Cellulose

85. In pyrrole,



the electron density is maximum on _____.

- (A) 2 and 5 (B) 2 and 3
 (C) 3 and 4 (D) 2 and 4

86. The solubility of $\text{AgCl}_{(s)}$ with solubility product 1.6×10^{-10} in 0.1 M NaCl solution would be _____.

- (A) zero (B) $1.26 \times 10^{-5} \text{ M}$
 (C) $1.6 \times 10^{-9} \text{ M}$ (D) $1.6 \times 10^{-11} \text{ M}$

87. The molar conductivity of a 0.5 mol/dm^3 solution of AgNO_3 with electrolytic conductivity of $5.76 \times 10^{-3} \text{ S cm}^{-1}$ at 298 K is _____.

- (A) $28.8 \text{ S cm}^2/\text{mol}$ (B) $2.88 \text{ S cm}^2/\text{mol}$
 (C) $11.52 \text{ S cm}^2/\text{mol}$ (D) $0.086 \text{ S cm}^2/\text{mol}$

88. The percentage of pyridine ($\text{C}_5\text{H}_5\text{N}$) that forms pyridinium ion $\text{C}_5\text{H}_5\text{NH}^+$ in a 0.10 M aqueous pyridine solution is _____.

- (A) 1.6 % (B) 0.0060 %
 (C) 0.013 % (D) 0.77 %

89. The coagulation values in millimoles per litre of the electrolytes used for the coagulation of As_2S_3 are given below:

- I. (NaCl) = 52 II. (BaCl_2) = 0.69
 III. (MgSO_4) = 0.22

The CORRECT order of their coagulating power is _____.

- (A) $\text{III} > \text{I} > \text{II}$ (B) $\text{I} > \text{II} > \text{III}$
 (C) $\text{II} > \text{I} > \text{III}$ (D) $\text{III} > \text{II} > \text{I}$

90. The suspension of slaked lime in water is known as _____.

- (A) aqueous solution of slaked lime
 (B) lime water
 (C) quick lime
 (D) milk of lime



91. Study the four statements (i - iv) given below and select the two correct ones out of them.
- Definition of biological species was given by Ernst Mayr.
 - Photoperiod does not affect reproduction in plants.
 - Binomial nomenclature system was given by R. H. Whittaker.
 - In unicellular organisms, reproduction is synonymous with growth.

The two correct statements are

- (A) i and ii (B) ii and iii
(C) iii and iv (D) i and iv
92. Which one of the following statements is not correct?
- Water Hyacinth growing in the standing water, drains oxygen from water that leads to the death of fishes.
 - Offspring produced by the asexual reproduction are called clone.
 - Microscopic, motile, asexual reproductive structures are called zoospores.
 - In potato, banana and ginger, the plantlets arise from the internodes which are present in the modified stem.

93. Which one of the following is WRONG for fungi?
- They are both unicellular and multicellular.
 - They are eukaryotic.
 - All fungi possess a purely cellulosic cell wall.
 - They are heterotrophic.

94. The ovule of an angiosperm is technically equivalent to
- megaspore
 - megasporangium
 - megasporophyll
 - megaspore mother cell

95. How many plants among *Indigofera*, *Sesbania*, *Salvia*, *Allium*, *Aloe*, Mustard, Groundnut, Radish, Gram, and Turnip have stamens with different lengths in their flowers?
- Six
 - Three
 - Four
 - Five

96. Which of the following is hormone releasing IUD?
- Cu7
 - LNG – 20
 - Multiload 375
 - Lippes loop

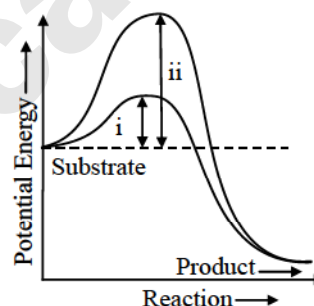
97. Choose the correct statement.
- All mammals are viviparous.
 - All cyclostomes do not possess jaws and paired fins.
 - All reptiles have a three-chambered heart.
 - All pisces have gills covered by an operculum.

98. A true breeding plant is
- always homozygous recessive in its genetic constitution
 - one that is able to bred on its own
 - produced due to cross pollination among unrelated plants
 - near homozygous and produces offsprings of its kind

99. Smooth muscles are
- voluntary, spindle-shaped, uninucleate
 - involuntary, fusiform, non-striated
 - voluntary, multinucleate, cylindrical
 - involuntary, cylindrical, striated

100. Genetic drift operates in
- slow reproductive population
 - small isolated population
 - large isolated population
 - non-reproductive population

101. Which of the following describes the given graph correctly?



- Exothermic reaction with energy i in absence of enzyme and ii in presence of enzyme.
- Endothermic reaction with energy i in presence of enzyme and ii in absence of enzyme.
- Exothermic reaction with energy i in presence of enzyme and ii in absence of enzyme.
- Endothermic reaction with energy i in absence of enzyme and ii in presence of enzyme.

102. Match Column-I with Column-II and select the correct option using the codes given below:

	Column-I		Column-II
i.	Citric acid	a.	<i>Trichoderma</i>
ii.	Cyclosporin A	b.	<i>Clostridium</i>
iii.	Statins	c.	<i>Aspergillus</i>
iv.	Butyric acid	d.	<i>Monascus</i>

	i.	ii.	iii.	iv.
(A)	c.	d.	a.	b.
(B)	a.	c.	b.	d.
(C)	c.	a.	d.	b.
(D)	a.	d.	b.	c.



103. Oxidative phosphorylation is
 (A) formation of ATP by energy released from electrons removed during substrate oxidation.
 (B) formation of ATP by transfer of phosphate group from a substrate to ADP.
 (C) oxidation of phosphate group in ATP
 (D) addition of phosphate group to ATP
104. Which kind of therapy was given in 1990 to a four-year-old girl with adenosine deaminase (ADA) deficiency?
 (A) Radiation therapy (B) Gene therapy
 (C) Chemotherapy (D) Immunotherapy
105. Lungs do not collapse between breaths and some air always remains in the lungs which can never be expelled because
 (A) pressure in the lungs is higher than the atmospheric pressure.
 (B) there is a negative pressure in the lungs.
 (C) there is a negative intrapleural pressure pulling at the lung walls.
 (D) there is a positive intrapleural pressure.
106. Which of the following National Parks is home to the famous musk deer or hangul?
 (A) Dachigam National Park, Jammu and Kashmir
 (B) Keibul Lamjao National Park, Manipur
 (C) Bandhavgarh National Park, Madhya Pradesh
 (D) Eagle nest Wildlife Sanctuary, Arunachal Pradesh
107. Name the blood cells, whose reduction in number can cause clotting disorder, leading to excessive loss of blood from the body.
 (A) Thrombocytes (B) Erythrocytes
 (C) Leucocytes (D) Neutrophils
108. The highest DDT concentration in aquatic food chain shall occur in
 (A) eel (B) phytoplankton
 (C) seagull (D) crab
109. Name the ion responsible for unmasking of active sites for myosin for cross-bridge activity during muscle contraction.
 (A) Potassium (B) Calcium
 (C) Magnesium (D) Sodium
110. Pollination in water hyacinth and water lily is brought about by the agency of
 (A) bats (B) water
 (C) insects or wind (D) birds
111. Match the Column-I with Column-II for housefly classification and select the correct option using the codes given below:

	Column I		Column II
i.	Family	a.	Diptera
ii.	Order	b.	Arthropoda

iii.	Class	c.	Muscidae
iv.	Phylum	d.	Insecta

- (A) i – d, ii – b, iii – a, iv – c
 (B) i – c, ii – a, iii – d, iv – b
 (C) i – b, ii – c, iii – d, iv – a
 (D) i – d, ii – c, iii – b, iv – a
112. Which of the following depicts the correct pathway of transport of sperms?
 (A) Efferent ductules → Rete testis → Vas deferens → Epididymis
 (B) Rete testis → Efferent ductules → Epididymis → Vas deferens
 (C) Rete testis → Epididymis → Efferent ductules → Vas deferens
 (D) Rete testis → Vas deferens → Efferent ductules → Epididymis
113. Free-central placentation is found in
 (A) *Citrus* (B) *Dianthus*
 (C) *Argemone* (D) *Brassica*
114. If a colour-blind man marries a woman who is homozygous for normal colour vision, the probability of their son being colour-blind is
 (A) 1 (B) 0 (C) 0.5 (D) 0.75
115. The balloon-shaped structures called tyloses
 (A) are linked to the ascent of sap through xylem vessels
 (B) originate in the lumen of vessels
 (C) characterize the sapwood
 (D) are extensions of xylem parenchyma cells into vessels
116. A molecule that can act as a genetic material must fulfill the traits given below, except
 (A) It should be able to express itself in the form of 'Mendelian characters'.
 (B) It should be able to generate its replica.
 (C) It should be unstable structurally and chemically.
 (D) It should provide the scope for slow changes that are required for evolution.
117. When cell has stalled DNA replication fork, which checkpoint should be predominantly activated?
 (A) Both G₂/M and M (B) G₁/S
 (C) G₂/M (D) M
118. Which of the following sets of diseases is caused by bacteria?
 (A) Herpes and influenza
 (B) Cholera and tetanus
 (C) Typhoid and smallpox
 (D) Tetanus and mumps
119. Phytochrome is a
 (A) chromoprotein (B) flavoprotein
 (C) glycoprotein (4) lipoprotein



120. Which of the following restriction enzymes produces blunt ends?
 (A) *Hind* III (B) *Sal* I
 (C) *Eco*RV (D) *Xho* I
121. You are given a tissue with its potential for differentiation in an artificial culture. Which of the following pairs of hormones would you add to the medium to secure shoots as well as roots?
 (A) Gibberellin and abscisic acid
 (B) IAA and gibberellin
 (C) Auxin and cytokinin
 (D) Auxin and abscisic acid
122. Red List contains data or information on
 (A) marine vertebrates only
 (B) all economically important plants
 (C) plants whose products are in international trade
 (D) threatened species
123. The posterior pituitary gland is NOT a 'true' endocrine gland because
 (A) it secretes enzymes
 (B) it is provided with a duct
 (C) it only stores and releases hormones
 (D) it is under the regulation of hypothalamus
124. A lake which is rich in organic waste may result in
 (A) mortality of fish due to lack of oxygen
 (B) increased population of aquatic organisms due to minerals
 (C) drying of the lake due to algal bloom
 (D) increased population of fish due to lots of nutrients
125. The label of a herbarium sheet does not carry information on
 (A) height of the plant
 (B) date of collection
 (C) name of collector
 (D) local names
126. Which one of the following generates new genetic combinations leading to variation?
 (A) Nucellar polyembryony
 (B) Vegetative reproduction
 (C) Parthenogenesis
 (D) Sexual reproduction
127. Select the WRONG statement.
 (A) Diatoms are microscopic and float passively in water.
 (B) The walls of diatoms are easily destructible.
 (C) 'Diatomaceous earth' is formed by the cell walls of diatoms.
 (D) Diatoms are chief producers in the oceans.
128. In majority of angiosperms
 (A) a small central cell is present in the embryo sac
 (B) egg has a filiform apparatus
 (C) there are numerous antipodal cells
 (D) reduction division occurs in the megaspore mother cells
129. The term 'polyadelphous' is related to
 (A) calyx (B) gynoecium
 (C) androecium (D) corolla
130. Match Column I with Column II and select the correct option using the codes given below:
- | | Column I | | Column II |
|------|---------------|----|---------------------------|
| i. | Mons pubis | a. | Embryo formation |
| ii. | Antrum | b. | Sperm |
| iii. | Trophectoderm | c. | Female external genitalia |
| iv. | Nebenkern | d. | Graafian follicle |
-
- | | i | ii | iii | iv |
|-----|---|----|-----|----|
| (A) | a | d | c | b |
| (B) | c | d | b | a |
| (C) | c | d | a | b |
| (D) | c | a | d | b |
131. In male cockroaches, sperms are stored in which part of the reproductive system?
 (A) Vas deferens (B) Seminal vesicles
 (C) Mushroom glands (D) Testes
132. Embryo with more than 16 blastomeres formed due to *in vitro* fertilization is transferred into
 (A) cervix (B) uterus
 (C) fallopian tube (D) fimbriae
133. Which of the following is least likely to be involved in stabilizing the three-dimensional folding of most proteins?
 (A) Ester bonds
 (B) Hydrogen bonds
 (C) Electrostatic interaction
 (D) Hydrophobic interaction
134. Which of the following rRNAs acts as structural RNA as well as ribozyme in bacterial?
 (A) 5.8 S rRNA (B) 5 S rRNA
 (C) 18 S rRNA (D) 23 S rRNA
135. During cell growth, DNA synthesis takes place in
 (A) M phase (B) S phase
 (C) G₁ phase (D) G₂ phase
136. Which of the following is the CORRECT sequence of events in the origin of life?
 I. Formation of protobionts
 II. Synthesis of organic monomers
 III. Synthesis of organic polymers
 IV. Formation of DNA-based genetic systems
 (A) II, III, IV, I (B) II, III, IV
 (C) I, III, II, IV (D) II, III, I, IV
137. Which is essential for the growth of root tip?
 (A) Mn (B) Zn
 (C) Fe (D) Ca



138. Which of the following is correct regarding AIDS causative agent HIV?
- (A) HIV is enveloped virus containing one molecule of single-stranded RNA and one molecule of reverse transcriptase.
(B) HIV is enveloped virus that contains two identical molecules of single-stranded RNA and two molecules of reverse transcriptase.
(C) HIV is unenveloped retrovirus.
(D) HIV does not escape but attacks the acquired immune response.
139. Which hormones do stimulate the production of pancreatic juice and bicarbonate?
- (A) Insulin and glucagon
(B) Angiotensin and epinephrine
(C) Gastrin and insulin
(D) Cholecystokinin and secretin
140. Among the following edible fishes, which one is a marine fish having rich source of omega-3 fatty acids?
- (A) Mackerel (B) Mystus
(C) Magur (D) Mrigala
141. Name a peptide hormone which acts mainly on hepatocytes, adipocytes and enhances cellular glucose uptake and utilization.
- (A) Gastrin (B) Insulin
(C) Glucagon (D) Secretin
142. Which of the following is CORRECT for *r*-selected species?
- (A) Small number of progeny with large size
(B) Large number of progeny with small size
(C) Large number of progeny with large size
(D) Small number of progeny with small size
143. Methanogens belongs to
- (A) Eubacteria (B) Archaeobacteria
(C) Dinoflagellates (D) Slime moulds
144. The mechanism that causes a gene to move from one linkage group to another is called
- (A) inversion (B) duplication
(C) translocation (D) crossing-over
145. Which of the metabolites / biomolecules is common to respiration mediated breakdown of fats, carbohydrates and proteins?
- (A) Fructose-1, 6-bisphosphate
(B) Pyruvic acid
(C) Acetyl-CoA
(D) Glucose-6-phosphate
146. In Hardy-Weinberg equation, the frequency of heterozygous individual is represented by
- (A) q^2 (B) p^2
(C) $2pq$ (D) pq
147. A few drops of sap were collected by cutting across a plant stem by a suitable method. The sap was tested chemically. Which one of the following test results indicates that it is phloem sap?
- (A) Acidic
(B) Alkaline
(C) Low refractive index
(D) Absence of sugar
148. Several hormones like hCG, hPL, estrogen, progesterone are produced by
- (A) pituitary (B) ovary
(C) placenta (D) fallopian tube
149. The part of nephron involved in active reabsorption of sodium is
- (A) descending limb of Henle's loop
(B) distal convoluted tubule
(C) proximal convoluted tubule
(D) Bowman's capsule
150. Which of the following is NOT a component of downstream processing?
- (A) Expression (B) Separation
(C) Purification (D) Preservation
151. Choose the CORRECT statement.
- (A) Receptors do not produce graded potentials.
(B) Nociceptors respond to changes in pressure.
(C) Meissner's corpuscles are thermoreceptors.
(D) Photoreceptors in the human eye are depolarized during darkness and become hyperpolarized in response to the light stimulus.
152. Which of the following is incorrect regarding vasectomy?
- (A) No sperm occurs in seminal fluid.
(B) No sperm occurs in epididymis.
(C) Vasa deferentia is cut and tied.
(D) Irreversible sterility.
153. The process which makes major difference between C_3 and C_4 plants is
- (A) respiration
(B) glycolysis
(C) Calvin cycle
(D) photorespiration
154. Taylor conducted the experiments to prove semi-conservative mode of chromosome replication on
- (A) *E. coli*
(B) *Vinca rosea*
(C) *Vicia faba*
(D) *Drosophila melanogaster*



155. Match the stages of meiosis in Column I to their characteristic features in Column II and select the correct option using the codes given below.

	Column I		Column II
i.	Pachytene	a.	Pairing of homologous chromosomes
ii.	Metaphase I	b.	Terminalization of chiasmata
iii.	Diakinesis	c.	Crossing over takes place
iv.	Zygotene	d.	Chromosomes align at equatorial plate

- (A) i-d, ii-c, iii-b, iv-a
 (B) i-c, ii-d, iii-b, iv-a
 (C) i-a, ii-d, iii-b, iv-c
 (D) i-b, ii-d, iii-c, iv-a
156. The chronological order of human evolution from early to the recent is
 (A) *Australopithecus* → *Homo habilis* → *Ramapithecus* → *Homo erectus*
 (B) *Australopithecus* → *Ramapithecus* → *Homo habilis* → *Homo erectus*
 (C) *Ramapithecus* → *Australopithecus* → *Homo habilis* → *Homo erectus*
 (D) *Ramapithecus* → *Homo habilis* → *Australopithecus* → *Homo erectus*
157. Serum differs from blood in
 (A) lacking antibodies
 (B) lacking globulins
 (C) lacking albumins
 (D) lacking clotting factors
158. How many hot spots of biodiversity in the world have been identified till date by Norman Myers?
 (A) 43 (B) 17
 (C) 25 (D) 34
159. The partial pressure of oxygen in the alveoli of the lungs is
 (A) less than that of carbon dioxide
 (B) equal to that in the blood
 (C) more than that in the blood
 (D) less than that in the blood
160. Biological Oxygen Demand (BOD) may not be good index for pollution for water bodies receiving effluents from
 (A) domestic sewage
 (B) dairy industry
 (C) petroleum industry
 (D) sugar industry
161. Which one of the following statements is WRONG?
 (A) *Laminaria* and *Sargassum* are used as food.
 (B) Algae increase the level of dissolved oxygen in the immediate environment.

- (C) Algin is obtained from red algae, and carrageenan from brown algae.
 (D) Agar-agar is obtained from *Gelidium* and *Gracilaria*.

162. Interspecific hybridisation is the mating of
 (A) more closely related individuals within same breed for 4 - 6 generations
 (B) animals within same breed without having common ancestors
 (C) two different related species
 (D) superior males and females of different breeds

163. Match Column I with column II and select the correct option using the codes given below.

	Column I		Column II
i.	Pistils fused together	a.	Gametogenesis
ii.	Formation of gametes	b.	Pistillate
iii.	Hyphae of higher Ascomycetes	c.	Syncarpous
iv.	Unisexual female flower	d.	Dikaryotic

- (A) i-d, ii-c, iii-a, iv-b
 (B) i-b, ii-a, iii-d, iv-c
 (C) i-a, ii-b, iii-d, iv-c
 (D) i-c, ii-a, iii-d, iv-b
164. The primary producers of the deep-sea hydrothermal vent ecosystem are
 (A) coral reefs
 (B) green algae
 (C) chemosynthetic bacteria
 (D) blue-green algae
165. Select the WRONG statement.
 (A) Mycoplasma is a wall-less micro-organism.
 (B) Bacterial cell wall is made up of peptidoglycan.
 (C) Pili and fimbriae are mainly involved in motility of bacterial cells.
 (D) Cyanobacteria lack flagellated cells.
166. DNA-dependent RNA polymerase catalyzes transcription on the strand of the DNA which is called the
 (A) antistrand (B) template strand
 (C) coding strand (D) alpha strand
167. Conifers are adapted to tolerate extreme environmental conditions because of
 (A) broad hardy leaves
 (B) superficial stomata
 (C) thick cuticle
 (D) presence of vessels



168. Osteoporosis, an age related disease of skeletal system, may occur due to
(A) accumulation of uric acid leading to inflammation of joints.
(B) immune disorder affecting neuromuscular junction leading to fatigue.
(C) high concentration of Ca^{++} and Na^+
(D) decreased level of estrogen
169. A foreign DNA and plasmid cut by the same restriction endonuclease can be joined to form a recombinant plasmid using
(A) ligase
(B) *EcoR* I
(C) *Taq* polymerase
(D) polymerase II
170. Select the mismatch.
(A) Methanogens – Prokaryotes
(B) Gas vacuoles – Green bacteria
(C) Large central vacuoles – Animal cells
(D) Protists – Eukaryotes
171. The principle of competitive exclusion was stated by
(A) Verhulst and Pearl (B) C. Darwin
(C) G.F. Gause (D) MacArthur
172. Radial symmetry is found in the flowers of
(A) *Cassia* (B) *Brassica*
(C) *Trifolium* (D) *Pisum*
173. The equivalent of a structural gene is
(A) recon (B) muton
(C) cistron (D) operon
174. A cell organelle containing hydrolytic enzymes is
(A) mesosome (B) lysosome
(C) microsome (D) ribosome
175. Which of the following is correctly matched?
(A) Stratification – Population
(B) Aerenchyma – *Opuntia*
(C) Age pyramid – Biome
(D) *Parthenium hysterophorus* – Threat to biodiversity
176. Cortex is the region found between
(A) endodermis and vascular bundle
(B) epidermis and stele
(C) pericycle and endodermis
(D) endodermis and pith
177. If '+' sign is assigned to beneficial interaction, '-' sign to detrimental and '0' sign to neutral interaction, then the population interaction represented by '+' '-' refers to
(A) parasitism (B) mutualism
(C) amensalism (D) commensalism
178. A non-proteinaceous enzyme is
(A) deoxyribonuclease
(B) lysozyme
(C) ribozyme
(D) ligase
179. Stirred-tank bioreactors have been designed for
(A) addition of preservatives to the product
(B) purification of the product
(C) ensuring anaerobic conditions in the culture vessel
(D) availability of oxygen throughout the process
180. Graves' disease is caused due to
(A) hypersecretion of adrenal gland
(B) hyposecretion of thyroid gland
(C) hypersecretion of thyroid gland
(D) hyposecretion of adrenal gland